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Dispatching, Positioning and Routing Resources for Wildfire Initial Attack

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ABSTRACT

Wildfire is a global issue that requires contributions from different fields to address its multiple and potentially severe impacts. The subject of this paper is wildfire suppression. In particular, a mixed integer programming (MIP) model is formulated for dispatching, positioning, and routing firefighting resources (e.g., crews and helicopters) in the initial attack. Following the minimum travel time principle, wildfire spread is modeled as the shortest path in a network, as is common in the literature for this type of approach. As for modeling resource movements and attack positions, it is proposed that each type of resource has its own network (e.g., roads) and interaction with the fire spread network. A heuristic, which alternates between MIP (for dispatching and positioning) and shortest path algorithms (for routing) is also conceived, and shown to provide significantly better solutions than solving the complete MIP model. Experiments with data from an actual landscape in Portugal were conducted with pyO3F (a Python framework being developed under the scope of the project “An optimization framework for reducing forest fire”). pyO3F builds the networks of fire and resources from vector and raster files (e.g., land use, altitudes, roads) and parameters from the user (e.g., resolution, fire behavior model). After an optimization approach is applied to a given scenario (characterized by a set of ignition nodes and a wind direction and intensity), pyO3F returns results in text (e.g., the burned area in given instants) and vector files (e.g., fire spread, resources movements, and attacks).

1 | Introduction

Wildfire is a global major issue that needs to be addressed at several decision levels and building on the intersection of different fields. It may have many negative impacts, including environmental and economic, but, most of all, it is a direct threat to human lives. For example, in the 2017 fires in Portugal [1], 2021 fires in the Mediterranean [2] and in the 2022 fires in Maui, USA, [3], 117, more than 100 and at least 114 lives were taken, respectively.

In this paper, we consider the operational problem that occurs after a fire is detected and assessed. The problem consists of deciding which firefighting resources (e.g., crews and helicopters) should be dispatched and where they should attack the fire.

As in previous approaches from the literature, we represent the landscape as a network where nodes are associated with cells and arcs are associated with potential fire transmission between two adjacent cells. Fire spread is modeled through the Minimum Travel Time (MTT) principle [4] that states that, when

represented in a network, the fire path between any two nodes in a landscape is the quickest path with respect to the fire transmission times of the arcs.

Modeling fire spread with an approach based on MTT allows the development of approaches that integrate fire spread and resource activities in the same model (it is an approach of the integrative type as classified in Reference [5]). The simplest way of achieving this integration is to use the network representing the landscape to represent both fire spread and resource positions.

We contribute with a more elaborated (and closer to reality) way of integration by defining the concept of the type of resource and associating with each type of resource a specific network (e.g., roads) and specific attributes related to its movement, attack positions, and interaction with fire. The fire spread and resource activities can thus be seen as occurring in a layered network with a layer for fire, a layer for each resource type, and besides the arcs of each layer, arcs between nodes of the resource layers and nodes of the fire layer (representing suppression activities or safety impositions). This approach is relevant for achieving a model closer to reality (fire spread and resource movements are too different to be represented in the same network) but also to make the conceptualization and implementation of fire-related optimization models easier in general.

A second contribution of this work is to explicitly model the route that each resource unit takes from its base to its attack position and its relation with fire spread. The first consideration is that the duration of the route of a resource is relevant since it must arrive at the attack position before the fire. A second consideration is that, for some resources, also the intermediate positions of the route must be crossed before fire. That is the case of *terrestrial* resources (also termed ground resources), as opposed to *aerial* resources where the intermediate positions are not conditioned by fire.

We formulate a mixed integer programming model (MIP) with lexicographic objectives: The first one is to minimize the burned area at a given instant, the second one is to minimize the dispatch cost (i.e., the cost of the resources effectively used), the third one, is to minimize the duration of the routes. The first objective can be seen as an indicator of the level of success of the initial attack. The initial attack (in Portugal, with a maximum duration of 90 min after the fire alert [6]) is followed by an extended attack (if fire was not extinguished), which has a different nature (e.g., additional resources from outside the region being requested). In this study, we confine our problem to the first T units of time (in the experiments, we set T to 90 min).

To obtain solutions to larger instances than the ones obtained with the MIP model and within a reasonable time interval (i.e., few minutes in an effective dispatch), as a third contribution, we developed a heuristic, termed Positioning and Routing Heuristic (PRH), that iterates between a MIP that selects which resources to dispatch and their attack positions, and a modified Dijkstra's algorithm to obtain, for each resource unit, the quickest route to its attack position that avoids fire.

To demonstrate the practical potential of the proposed approach, it was implemented using pyO3F (a Python framework being

developed under the scope of the project “An optimization framework for reducing forest fire”). In this framework, fuel, altitudes, and different ignitions and winds are used with fire behavior models to generate scenarios in an actual landscape. Actual roads are used to define the network of terrestrial resources. Interaction with the user, including results on fire spread, resource routes, and attack positions is mainly done by vector files. A relevant feature of pyO3F is that the user can select the resolution of the landscape, controlling the level of detail of the models and the size of the networks used in the optimization models.

The rest of this paper is organized as follows. In the next section, Section 2, we contextualize the problem addressed in the scope of fire management problems suitable for optimization approaches. We provide an overview of optimization approaches to fire-related decision problems. In Section 3, we formally define the problem and introduce the notation used. In Section 4 we present the proposed MIP model and in the following section, Section 5, the PRH. We present and discuss the computational experiments in Section 6 and the conclusions are summarized in Section 7, which also includes some issues that point out some research directions.

2 | Literature Review

2.1 | Fire Management and Optimization: Overview and a Typology

We begin this section by providing an overview of the most representative fire problems where optimization has been applied in fire management. Optimization and other Operational Research (OR) approaches have been surveyed in Reference [7] (reviewing contributions of Operations Research to fire management since its beginnings-the late 1950s/early 1960s), [8], [9], and [10].

Approaches to decision-making in fire management have been organized differently. For example [8], considers prevention, detection, management of fire resources (further divided into strategic, daily deployment, initial attack dispatch), containment, and large fire management (initial attack and large fire management). In Reference [11], examples of optimization problems are framed in the four groups of activities in disaster response [12]: Mitigation, preparedness, response, recovery. Focusing on fire suppression, [9] considers two major groups, preparedness, and response, each with several levels and factors; the same division is used in Reference [13]; other authors divide the activities in pre suppression and suppression.

We propose a typology for optimization problems related to wild-fires based on two dimensions: The length of the time horizon and the presence of an actual fire. The length of the time horizon is divided into short (S) or medium-long (M). The presence of an actual fire, that is, an ignition occurred in the landscape and was detected, is represented by F and the absence by N (no fire). In Table 1, several problems are typified. These can be seen as base problems that can be integrated. For example, deployment and dispatch can be integrated (e.g., in a two-stage stochastic approach where the first stage is related to deployment and the second stage to dispatch in different scenarios-e.g., different ignitions).

TABLE 1 | Types of optimization problems in wildfire.

		Presence of fire	
		No fire	Fire
Length of the time horizon	Short (e.g., day)	Deployment-Allocation/reallocation to/between bases Scheduling and routing of detection resources Prepositioning	Dispatch-Allocation of resources to a fire Positioning and/or routing for fire suppression Routing for asset protection Evacuation
	Medium-large (e.g., year)	Fuel management Location of fixed equipment for detection Resources basing Landscape planning Budgeting	—

In this paper, we address a dispatch with positioning and routing problem (a problem of type SF). The dispatch refers to the fact that our approach suggests which resources, among the available ones (of different types), should be allocated to attack fire. It further details where each resource should be positioned and the shortest length route that avoids fire.

2.2 | Related Works

The problem under study can be seen as an extension of the positioning problem to include routing resources in their own networks. The (deterministic) positioning problem has been addressed before in Reference [5, 14–18]. The last two references showed that it is possible to reach good quality solutions in reasonable amounts of time with iterated local search and logic-based benders decomposition, respectively. The positioning problem with uncertainty on the wind is addressed in Reference [19]. In Reference [20], a robust optimization approach is described for the problem of positioning resources with uncertainty on the number of resources needed to suppress fire at a node. All these approaches are based on the MTT principle.

With respect to these works, we contribute to lexicographic optimization and the definition of layered networks for different resource types. One main contribution is modeling and solving an extension of the positioning problem that takes into account the routing of the resources in an initial attack setting.

Routing has been considered in minimizing the total time of resources usage in fire suppression [21], in the base asset protection problem [22] or in the integration of asset protection with location and allocation under uncertainty [11, 23].

A major difference between those works related to routing and this one is the type of approach. Using the classification of [5], those approaches are of the sequential type: Fire is used in the optimization models through parameters that are not

influenced by the resources. Our approach (as the ones based on the MTT principle mentioned before) is integrative: in a single (MIP) model, fire spread influences the feasible movements and attack positions of the resources, and the attack positions of the resources influence how fire spreads. A second major difference is that, in this paper (as in the MTT papers mentioned before), only one position attack is considered per resource unit, and in graph terms, a route corresponds to a path. This is aligned with the subject of the paper—the initial attack which has a short duration.

A combination of the MTT principle for fire spread and resource movements is achieved in Reference [24], where the MIP model includes fireline construction constraints and uncertainty. Results on a small instance (11 by 12 grid cells) are provided.

3 | Problem Definition and Notation

The problem addressed in this paper is defined within a landscape where a wildfire has been detected. A set of firefighting resources of different types are available and it is intended to minimize the impact of fire, measured after a given period of time, by positioning the resources in locations where the fire is attacked. Besides positioning the resources, it is also intended to safely route them from their bases to their attack positions.

In our approach, the landscape is divided into hexagonal grid cells, and time is continuous. A layered network is used to represent the spatial dimension of the problem: One layer for fire spread and one layer for each resource type. In the fire layer, each node represents a cell, and each arc represents the potential transmission of fire between two adjacent nodes. In a resource-type layer, each node represents a location, and each arc represents the potential movement between locations. The time dimension is represented by continuous variables. We now introduce the notation used and detail some aspects of the problem definition, starting with the fire layer.

3.1 | Fire

Entities and parameters related to fire have superscript f and are the following:

- N^f set of nodes of the fire layer;
- I^f set of initial state nodes, $I^f \subset N^f$;
- A^f set of arcs of the fire network;
- m_{ij}^f fire transmission time between adjacent nodes i and j , $ij \in A^f$.

The set I^f characterizes the fire's initial state. A node belonging to I^f is in the fire perimeter, that is, it was reached by fire and is the origin of, at least, one arc to an unburned node. The set N^f (and thus I^f) does not include nodes that were reached by fire before the alert and have no outgoing arcs to unburned nodes. These definitions allow us to model any type of fire, including multiple ignitions and the presence of more than one fire.

3.2 | Resources

The set of resource types R is indexed by r . The nodes of the layer of a resource type are associated with potential attack positions and relevant locations to characterize potential movements (e.g., crossroads) which are represented by arcs.

We consider two classes of resources: Aerial and terrestrial. They are characterized in the same manner (with a network layer and parameters, see below). In opposition to aerial resources, terrestrial resources cannot be near burning locations, except to attack fire. The set of aerial resources is represented by S , $S \subseteq R$. The set of the terrestrial resources is then $R \setminus S$.

The entities and parameters of the resource type layers have superscript r and are the following:

- N^r set of nodes of the layer of resource type r , $r \in R$;
- A^r set of arcs of the layer of resource type r , $r \in R$;
- m_{pq}^r movement time between adjacent nodes p and q for resource type r , $r \in R$, $pq \in A^r$;

A resource type may have one or more units. The set of units of resource type r is represented by U^r . Units of the same resource type use the same layer, have the same interaction with fire (sets L^{rf} and F^{fr} , see below) and have the same attributes: Base location, preparation time (i.e., the time to leave the base after fire detection), movement time in each arc (e.g., same speed because vehicles are of the same type) and cost. The notation used is the following:

- b^r number of available resources of type r , $r \in R$;
- l^r base node for type of resource r , $r \in R$, $l^r \in N^r$;
- s^r preparation time for a resource of type r , $r \in R$;
- c^r unit cost of a resource of type r , $r \in R$;

- h_k^r duration of the quickest path of a resource of type r from its base l^r to a node k .

3.3 | Linking Fire and Resources

There are two types of relations of resources with fire: Resources may suppress fire (suppression relation-a resource attacks fire) and terrestrial resources must avoid fire when they are not suppressing it (safety relation-a resource must avoid a fire). We first discuss the suppression relation and then the safety relation.

To model actual fire suppression, we consider that each attack position of each resource covers a set of fire nodes, that is, if placed in that location that set of fire nodes will not transmit fire. The covering sets may take into account local characteristics (such as the difficulty of the terrain for suppressing fire at a given node but not at its neighbor, e.g., due to slope).

We define a set L^{rf} , for each resource type r , made of ordered pairs, (p, i) , for short, pi , such that:

- the first element is a node of resource type r layer, that is, $p \in N^r$;
- the second element is a node of the fire network, $i \in N^f$;
- if a unit of resource r is positioned at p , fire will not spread from i .

Each element of L^{rf} can be seen as an arc linking a node in the resource layer to a node in the fire layer.

A straightforward use of the set L^{rf} is to characterize a resource type with a suppression radius d^r , $r \in R$. A pair pi belongs to L^{rf} if the distance (in some metric, e.g., Euclidean) between p and i is smaller than or equal to d^r . An illustration is provided in Figure 1. A resource can only be positioned at node 1199 if it arrives there earlier than fire at nodes 109, 98, and 86. If a resource is positioned at resource node 1199, fire will not spread from each of those fire nodes.

An obvious concern in fire suppression is the safety of the firefighters. To model that a terrestrial crew must keep a safe distance

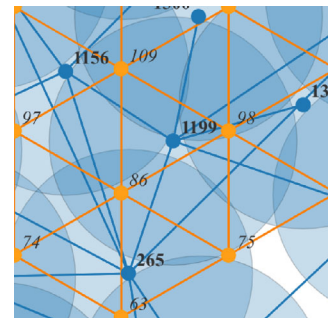


FIGURE 1 | Illustration of a suppression relation based on a radius. The fire layer is represented in orange with italic labels and a resource layer is represented in blue with bold labels. Suppression circles are also represented in blue.

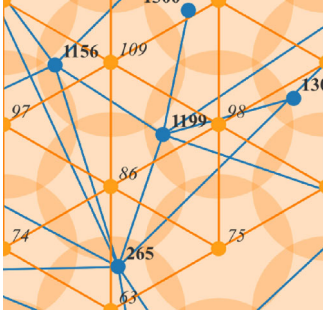


FIGURE 2 | Illustration of a safety relation based on a radius. The fire layer is represented in orange with italic labels and a resource layer is represented in blue with bold labels. Orange circles correspond to the unsafe area of each fire node for the given type of resource.

from the fire, and using the same approach as the one used with the fire suppression interaction, we define one set F^{fr} for each resource type r made of ordered pairs ip , such that the first element is a node of the fire network, $i \in N^f$, the second element is a node of the resource type r network, that is, $p \in N^r$. From the instant fire arrives at a node i , no terrestrial resource unit of type r can cross a node p such that $(i, p) \in F^{fr}$. Each element of F^{fr} can be seen as an arc linking a node in the fire layer to a node in the resource layer. As with set L^{fr} , elements of the sets F^{fr} can be defined by an unsafety radius e^r , $r \in R$. A pair ip belongs to F^{fr} if the distance between i and p is smaller than or equal to e^r . An illustration is provided in Figure 2. A resource can cross node 1199 only if it arrives there earlier than fire at nodes 98 and 86.

A feasible route of a resource is a path from its base to an attack node (in its resource type network) that satisfies the safety relation at all intermediate nodes and the suppression relation at the attack node.

3.4 | Objectives

We consider three hierarchic objectives: To minimize the burned area, minimize the cost of the resources used, and minimize the duration of the terrestrial routes. These objectives are aligned with the relevance of fire suppression by the negative impacts it has, but using only the necessary resources (many resources may be available) and, among them, the ones that can arrive faster to the attack position (in general, if many resources are available, the ones with a base closer to the attack position should be used).

4 | A MIP Model

In this section, we describe the proposed MIP model.

Fire spread is modeled by decision variables associated with the fire spread, that is, the fire arrival times at each node and the number of burned nodes at the evaluation instant, represented by T :

$$t_i^f - \text{instant of the fire arrival at node } i, \quad i \in N^f;$$

$$y_i^f = \begin{cases} 1, & \text{if node } i \text{ burns before instant } T \\ 0, & \text{otherwise} \end{cases}, \quad i \in N^f$$

The decisions where (i.e., in which nodes of the resource layer) the resources attack fire are defined differently based on the need to keep track of the time of each resource unit. For aerial resources, the arrival at the attack node must precede fire; therefore, the units can be aggregated in the definition of decision variables:

$$x_k^r = \begin{cases} 1, & \text{if a unit of resource of type } r \\ & \text{attacks fire at node } k \\ 0, & \text{otherwise} \end{cases}, \quad r \in S, k \in N^r$$

Terrestrial units must be treated individually because the feasibility of the route (not only the attack node) depends on how fire spreads. The definition of the decision variables for representing the nodes where terrestrial units attack is then:

$$x_p^{ru} = \begin{cases} 1, & \text{if unit } u \text{ of resource of type } r \\ & r \text{ attacks fire at node } p \\ 0, & \text{otherwise} \end{cases}, \quad r \in R \setminus S, u \in U^r, p \in N^r$$

To completely represent the route of each terrestrial unit, the following decision variables are defined:

$$t_p^{ru} - \text{instant when the resource unit } u \text{ arrives at node } p,$$

$$r \in R \setminus S, u \in U^r, p \in N^r;$$

$$w_{pq}^{ru} = \begin{cases} 1, & \text{if unit } u \text{ of resource of type } r \text{ uses arc } pq \\ 0, & \text{otherwise} \end{cases},$$

$$r \in R \setminus S, u \in U^r, pq \in A^r;$$

$$v_p^{ru} = \begin{cases} 1, & \text{if unit } u \text{ of resource of type } r \text{ crosses node } p \\ 0, & \text{otherwise} \end{cases},$$

$$r \in R \setminus S, u \in U^r, p \in N^r$$

We define sets $\delta^-(i)$ and $\delta^+(i)$ as the set of incoming and outgoing arcs of node i . The MIP model is the following.

$$\text{Lex} \quad \text{Min} \sum_{i \in N^f} y_i^f \quad (1)$$

$$\text{Min} \sum_{\substack{r \in R \setminus S, u \in U^r, \\ p \in N^r \setminus \{i^f\}}} c^r x_p^{ru} + \sum_{r \in S, k \in N^r \setminus \{i^f\}} c^r x_k^r \quad (2)$$

$$\text{Min} \sum_{\substack{r \in R \setminus S, u \in U^r, \\ pq \in A^r}} m_{pq}^r w_{pq}^{ru} \quad (3)$$

$$\text{Subject to:} \quad (4)$$

$$t_i^f = 0 \quad i \in I^f$$

$$t_j^f \leq t_i^f + m_{ij}^f + T \left(\sum_{\substack{r \in R \setminus S, u \in U^r, \\ p \in N^r \setminus \{i^f\}}} x_p^{ru} + \sum_{r \in S, k \in N^r \setminus \{i^f\}} x_k^r \right) \quad ij \in A^f \quad (5)$$

$$y_i^f \geq 1 - t_i^f / T \quad i \in N^f \quad (6)$$

$$t_p^{ru} = s^r \quad r \in R \setminus S, u \in U^r \quad (7)$$

$$\sum_{q \in \delta^-(i^f)} w_{iq}^{ru} = \sum_{p \in N^r} x_p^{ru} \quad r \in R \setminus S, u \in U^r \quad (8)$$

$$\sum_{q \in \delta^+(p)} w_{pq}^{ru} - \sum_{q \in \delta^-(p)} w_{qp}^{ru} = -x_p^{ru} \quad r \in R \setminus S, u \in U^r, p \in N^r \setminus \{l^r\} \quad (9)$$

$$\sum_{q \in \delta^+(l^r)} w_{l^r q}^{ru} \leq 1 \quad r \in R \setminus S, u \in U^r \quad (10)$$

$$t_q^{ru} \geq t_p^{ru} + m_{pq}^r - M^r(1 - w_{pq}^{ru}) \quad r \in R \setminus S, u \in U^r, pq \in A^r \quad (11)$$

$$v_p^{ru} = \sum_{q \in \delta^+(p)} w_{pq}^{ru} \quad r \in R \setminus S, u \in U^r, p \in N^r \quad (12)$$

$$t_p^{ru} \leq t_i^f + T(1 - v_p^{ru}) \quad i \in N^f, r \in R \setminus S, u \in U^r, ip \in F^{fr} \quad (13)$$

$$t_p^{ru} \leq t_i^f + T(1 - x_p^{ru}) \quad i \in N^f, r \in R \setminus S, u \in U^r, pi \in L^{rf} \quad (14)$$

$$\sum_{k \in N^r} x_k^r \leq b^r \quad r \in S \quad (15)$$

$$h_k^r x_k^r \leq t_i^f \quad i \in N^f, r \in S, ki \in L^{rf} \quad (16)$$

$$t_i^f \geq 0 \quad i \in N^f \quad (17)$$

$$y_i^f \in \{0, 1\} \quad i \in N^f \quad (18)$$

$$x_p^{ru} \in \{0, 1\} \quad r \in R \setminus S, u \in U^r, p \in N^r \quad (19)$$

$$t_p^{ru} \geq 0 \quad r \in R \setminus S, u \in U^r, p \in N^r \quad (20)$$

$$w_{pq}^{ru} \in \{0, 1\} \quad r \in R \setminus S, u \in U^r, pq \in A^r \quad (21)$$

$$v_p^{ru} \in \{0, 1\} \quad r \in R \setminus S, u \in U^r, p \in N^r \quad (22)$$

$$x_k^r \in \{0, 1\} \quad r \in S, k \in N^r \quad (23)$$

The first lexicographic objective function (1) minimizes the number of burned nodes, or equivalently, the burned area, until instant T . The second lexicographic objective function (2) minimizes the cost of the used resources. In this way, a resource is used only if effective, that is, if it contributes to the reduction of the burned area. The third objective chooses the quickest route for terrestrial resources (the quickest path of the aerial resources is defined by the suppression node and does not require optimization).

Constraints (4–6) refer to fire. Constraints (4) define the nodes where the fire is active when the alert occurs. Constraints (5) state that the fire arrival time at a node corresponds to the quickest way fire can arrive at the node (necessarily, from one of its neighbors). Given the first objective function and its sense, constraints (5) can be seen as a linearization of $t_j^f = \min\{t_i^f + m_{ij}^f + Tx : i \in \delta^-(j)\}$, $j \in N^f$, where $\delta^-(j)$ is the set of origin nodes of the arcs incoming at j and $x = \sum_{r \in R, u \in U^r, pi \in L^{rf}} x_p^{ru}$. Positioning a resource at a node forbids the fire transmission from all the fire nodes it covers to their neighbors. Therefore the evaluation instant of the number of burned nodes, T , can be used as the fire delay implied by the presence of the resource, assuring nodes with incoming

arcs from nodes with suppression will not burn within T time units. Constraints (6) relate the fire arrival time at a node with the node being burned before T .

Constraints (7–14) are related to terrestrial resources. For each resource type and unit, a route can only start after the preparation time (7). If suppression occurs at a node, a path from the base must exist ((8) and (9)). A unit can only be used once through (10). Constraints (11) define the arrival time of each unit of each resource at each node. The upper bound used, M^r is defined through (24):

$$M^r = \min \left\{ \sum_{pq \in A^r} m_{pq}^r, (|N^r| - 1) \max_{pq \in A^r} \{m_{pq}^r\} \right\} \quad (24)$$

To obtain a tighter upper bound, nodes with the quickest arrival times greater than the evaluation instant are removed, resulting in the set of arcs A^{rf} .

The purpose of constraints (12) is to identify the nodes that belong to the path of each terrestrial resource unit (with the exception of the last one, the suppression node). Constraints (13) force the resources to keep a safe distance from fire. The arrival of a resource unit ru at any node p of its path must precede the fire arrival at any node that covers i , $i \in F^{fr}$. Constraints (14) state that when a suppression occurs at a node, p , the resource unit must arrive earlier than fire in all nodes i where fire will be suppressed, that is, $i \in L^{rf}$. Otherwise, the resource would be used to suppress already-burned locations.

Constraints (15) ensure that the number of aerial resources does not exceed the available ones. For aerial resource units, times are relevant only to ensure that an attack node is reached before fire. This is accomplished by determining the earliest arrival time of each resource type r through a shortest path tree problem, on the resource network, rooted on the base of the resource type. Constraints (16) ensure that each aerial resource reaches the attack node before the fire.

The remaining constraints are relative to the domain of the variables.

In a preprocessing step, we attempt to reduce the dimension of the network(s) exploring the fact that what occurs beyond the evaluation instant does not affect the solution or the objective. For a layer, nodes where the quickest arrival time (from an initial fire perimeter in the case of the fire layer, or the base in the case of a resource-obtained from shortest path tree algorithms) exceeds the evaluation instant can be removed from the layer. It is relevant to note that the size of the network is influenced by the evaluation instant but also by the resolution. A small evaluation instant may not imply a small network if the resolution is high.

5 | Positioning and Routing Heuristic

In this section, we describe the proposed heuristic which is based on the decomposition of the problem in two subproblems (one for dispatching and positioning, PSP, and one for routing, RSP)

that are solved alternately until a feasible solution is obtained (which always happens). The PSP obtains attack positions for the resource units assuring their arrival *only* at the attack positions precedes the fire arrival. The RSP attempts to find *feasible* paths for the terrestrial resources between their bases and the attack positions provided by the PSP. A path is feasible if the arrival of the resource unit precedes the arrival of fire *in all* the nodes of the path. If, for a unit, no feasible path exists, the corresponding attack position is removed from the PSP and a new iteration begins. If there are feasible paths for all units, the heuristic stops with the current positions and paths.

5.1 | Positioning Subproblem

The PSP may be seen as resulting from treating the terrestrial resources as aerial and therefore it is much simpler (no time-related variables and constraints for the routes) and smaller (all variables are related to resource types and no need to have variables associated with units) than the one proposed in the previous Section. The path of the aerial resource units is determined by the solution of the PSP. The solution of the PSP also provides attack positions for the terrestrial resources. After an optimal solution is obtained, each attack position is associated with a unit.

The MIP model for the PSP is the following:

$$\begin{aligned}
 &\text{Lex} \\
 &\text{Min } \sum_{i \in N^f} y_i^f \\
 &\text{Min } \sum_{r \in R, k \in N^r \setminus \{I^r\}} c^r x_k^r \\
 &\text{Subject to:} \\
 &t_i^f = 0 \quad i \in I^f \\
 &t_j^f \leq t_i^f + m_{ij}^f + T \sum_{r \in R, ki \in L^f} x_k^r \quad ij \in A^f \\
 &y_i \geq 1 - t_i^f / T \quad i \in N^f \\
 &\sum_{k \in N^r} x_k^r \leq b^r \quad r \in R \\
 &h_k^r x_k^r \leq t_i^f \quad i \in N^f, r \in R, ki \in L^f \\
 &t_i^f \geq 0 \quad i \in N^f \\
 &y_i^f \in \{0, 1\} \quad i \in N^f \\
 &x_k^r \in \{0, 1\} \quad r \in R, k \in N^r
 \end{aligned}$$

The fire arrival times in an optimal solution of this model may not correspond to the ones associated with the shortest paths, disrespecting the MTT principle. This issue is discussed in Reference [14] and the references therein.

To overcome this issue and obtain the correct fire arrival times at all nodes a shortest path tree algorithm is applied taking into account the influence of the attack positions obtained with the MIP, that is, all outgoing arcs of fire nodes covered by a node with a resource are removed from the fire network. These fire

arrival times, obtained with the shortest path tree algorithm, are used in the next step of the approach, the RSP. We denote them as $\tilde{t}_i^f, i \in N^f$.

5.2 | Routing Subproblem

The RSP refers to the terrestrial resources and is decomposable by resource unit. The RSP of a resource unit consists of finding a feasible path for each terrestrial resource unit with upper bounds derived from the fire arrival times at the fire nodes. In this context, *feasible* means that the arrival time of the unit at each resource node of its path must precede the fire arrival time of each fire node within a given radius. For the attack node, a suppression radius applies, while, for the other nodes of the path, a safety radius applies.

Consider the subproblem of a unit u of a resource r positioned at node $\bar{k}$, that is, $x_{\bar{k}}^r = 1$ in the solution of the PSP.

The MIP model for a unit u with resource type r is the following. In the model below, (25) refers to the line immediately above (26)

$$\begin{aligned}
 &\text{Min } \sum_{pq \in A^r} m_{pq}^r w_{pq}^{ru} \\
 &\text{Subject to:} \\
 &t_{I^r}^{ru} = s^r \\
 &\sum_{q \in \delta^+(I^r)} w_{I^r q}^{ru} = 1 \\
 &\sum_{q \in \delta^+(p)} w_{pq}^{ru} - \sum_{q \in \delta^-(p)} w_{qp}^{ru} = 0 \quad p \in N^r \setminus \{I^r, \bar{k}\} \\
 &-\sum_{q \in \delta^-(\bar{k})} w_{q\bar{k}}^{ru} = -1 \\
 &t_q^{ru} \geq t_p^{ru} + m_{pq}^r - M^r(1 - w_{pq}^{ru}) \quad pq \in A^r \\
 &t_p^{ru} \leq \tilde{t}_i^f \quad i \in N^f, ip \in F^{f^r}, p \neq \bar{k} \quad (25) \\
 &t_{\bar{k}}^{ru} \leq \tilde{t}_{\bar{k}}^f \quad \bar{k}i \in L^f \quad (26) \\
 &t_p^{ru} \geq 0 \quad p \in N^r \\
 &w_{pq}^{ru} \in \{0, 1\} \quad pq \in A^r
 \end{aligned}$$

The RSP of one resource unit is therefore the shortest (quickest) path problem from its base to its attack position with upper bounds on the lengths (durations) of the nodes. A modification of Dijkstra's algorithm ([25, 26]), allows to solve optimally (and efficiently) this problem. The resource arrival time of a resource ru at node p (different from the attack node) cannot exceed (from (25)):

$$b_p^{ru} = \min_{i \in N^f, ip \in F^{f^r}, p \neq \bar{k}} \{\tilde{t}_i^f\} \quad (27)$$

And, for the attack node, (26) can be rewritten as:

$$b_{\bar{k}}^{ru} = \min_{\bar{k}i \in L^f} \{\tilde{t}_i^f\} \quad (28)$$

Constants b_p^{ru} and b_k^{ru} are the upper bounds that must be respected at each node of the resource network in the modified Dijkstra's algorithm.

The modification of Dijkstra's algorithm consists of an additional condition in the attempt to update the predecessor of a node. The original condition (when checking if p is a better predecessor of q than the current one) is

$$t_p^{ru} + m_{pq}^r < t_q^{ru} \quad (29)$$

For a node q other than the attack node, the original condition is replaced by:

$$t_p^{ru} + m_{pq}^r < t_q^{ru} \text{ and } t_p^{ru} + m_{pq}^r < b_q^{ru} \quad (30)$$

For the attack node, $\bar{k}$, the original condition is replaced by:

$$t_p^{ru} + m_{p\bar{k}}^r < t_{\bar{k}}^{ru} \text{ and } t_p^{ru} + m_{p\bar{k}}^r < b_{\bar{k}}^{ru} \quad (31)$$

If the problem is infeasible, the variable $x_{\bar{k}}^r$ is removed from the PSP of the next iteration. The algorithm stops when there is a feasible path for each attack position of each terrestrial resource.

5.3 | Algorithm

The algorithm of the proposed heuristic is formalized in Algorithm 1. MIP_{ini} stands for the initial MIP model of the PSP (the one of Subsection 5.1), while MIP_{cur} stands for the current MIP model of the PSP.

After the PSP initialization (line 1), the main cycle (lines 2 to 20) is repeated until a feasible solution is found.

The MIP optimization of line 3, obtains attack positions that are used when calculating the fire arrival times in line 4. The cycle of lines 6 to 20, initialized in line 5, is related to the RSP. For each terrestrial resource type, the cycle from lines 7 to 16 attempts to find the quickest feasible path (line 9) for each attack position (identified in line 8). If there is no feasible path for an attack position, the corresponding node is removed from the network of its resource (lines 11 to 19) and a new iteration begins with the updated MIP (line 3). This procedure is repeated until a feasible path (line 21) is found for all terrestrial resources.

6 | Computational Experiments

We implemented and tested the proposed approaches to assess their suitability for supporting decision-making in an initial attack setting. For a landscape of approximately 350 km², we obtained results on:

- how fire spreads in different scenarios (defined by the ignition location and wind);
- the influence of the resolution of the network in the ability to solve the problem(s);

ALGORITHM 1 | Positioning and routing heuristic.

```

1:  $MIP_{cur} \leftarrow MIP_{ini}$ 
2: repeat
3:   Optimize  $MIP_{cur}$ 
4:   Obtain fire arrival times at all nodes of the fire network
   with a shortest path tree algorithm
5:    $NoPath \leftarrow False$ 
6:   for  $r \in R \setminus S$  do
7:     for  $k \in N^r$  do
8:       if  $x_k^r == 1$  then
9:         Apply modified Dijkstra's algorithm between
         base  $l^r$  and attack node  $k$  with bounds obtained through (27)
         and (28)
10:        if no path from  $l^r$  to  $k$  then
11:          Remove  $x_k^r$  from  $MIP_{cur}$ 
12:           $NoPath \leftarrow True$ 
13:          Break
14:        end if
15:      end if
16:    end for
17:    if  $NoPath == True$  then
18:      Break
19:    end if
20:  end for
21: until  $NoPath == False$ 

```

- the effectiveness of different combinations of resources;
- how the feasibility of the solutions is affected when fire is not considered in the routes.

6.1 | Pyo3f and the Landscape

The proposed approaches have been implemented in pyO3F, a Python framework being developed under the scope of the project "An optimization framework for reducing forest fire". pyO3F uses *Geopandas*, *NetworkX* and, for optimization, *Gurobi*. All the computational tests were conducted on a machine with a processor Intel Core i7, 1 700 Mhz, 10 cores, 12 logical processors, and 32 GB of physical memory.

The region under study is located in the north of Portugal, as seen in Figure 3, near the municipalities of Ponte da Barca (in the top left corner) and Terras de Bouro (near the center), as well as the Peneda-Gerês National Park (on the right). The landscape measures are 20 889 m by 16 632 m.

In pyO3F, a set of parameters and two files are used as input to characterize the landscape. Figure 4 shows images of the two files: One with the altitudes and the other with the land use. The categorization of land use is based on the highest level (level 1) [27]. This level 1 categorization assigns one of nine classes to each area of the landscape: Artificialized territories, agriculture, pastures, agroforestry areas, forests, bushes, open spaces or with little vegetation, wetlands, and surface water bodies. Lower levels subdivide the upper-level classes, providing more detailed information.

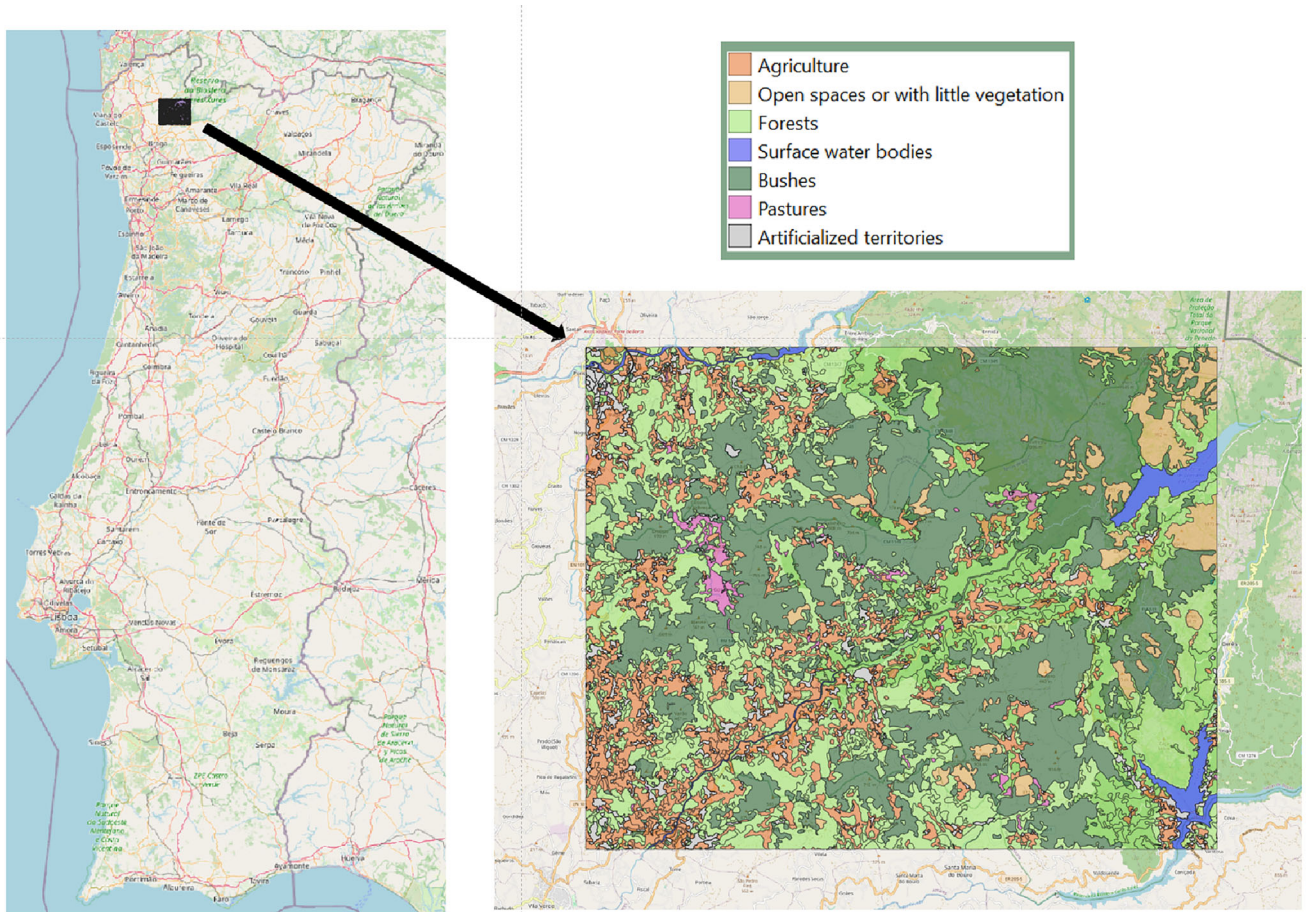


FIGURE 3 | Region under study.

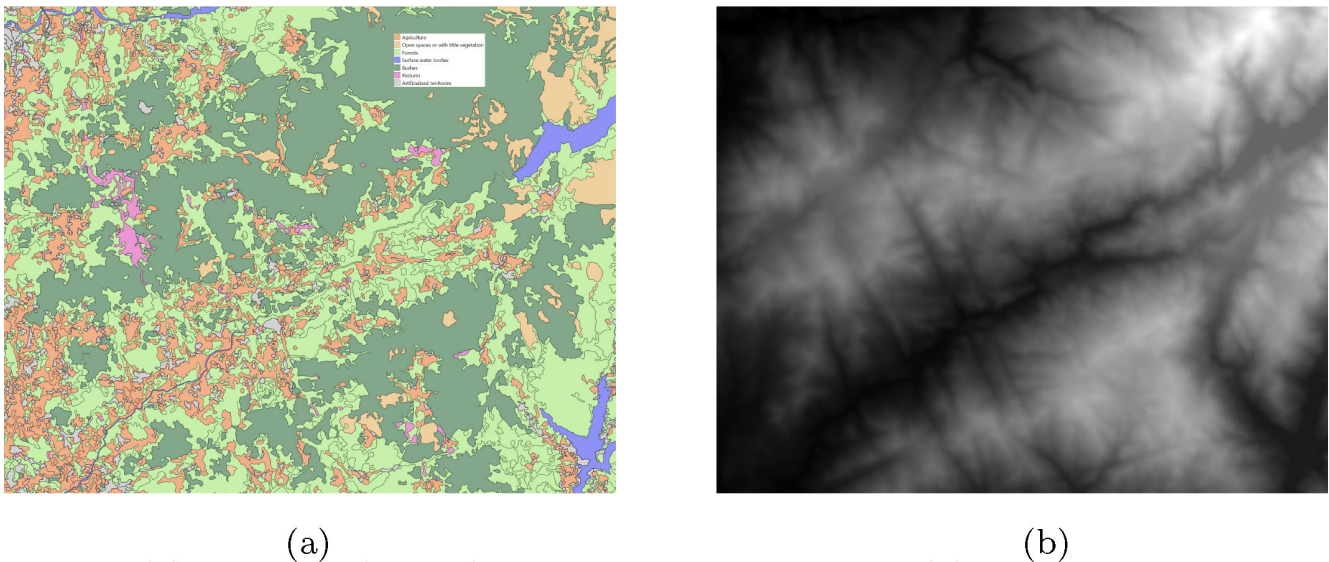


FIGURE 4 | Images of the input files for pyO3F. (a) Land use (level 1) and (b) Altitudes.

To allow modeling of the fire spread in the landscape, a network is constructed based on a hexagonal grid. The resolution (distance between hexagon centers) is set by an input parameter. Each node (the center of a hexagon) is associated with the land

use with the highest area within the hexagon. Unburnable nodes are removed from the network (except for nodes corresponding to the artificialized land use, as discussed below). In Figure 5, the hexagonal cells, nodes, and arcs of the fire network are presented

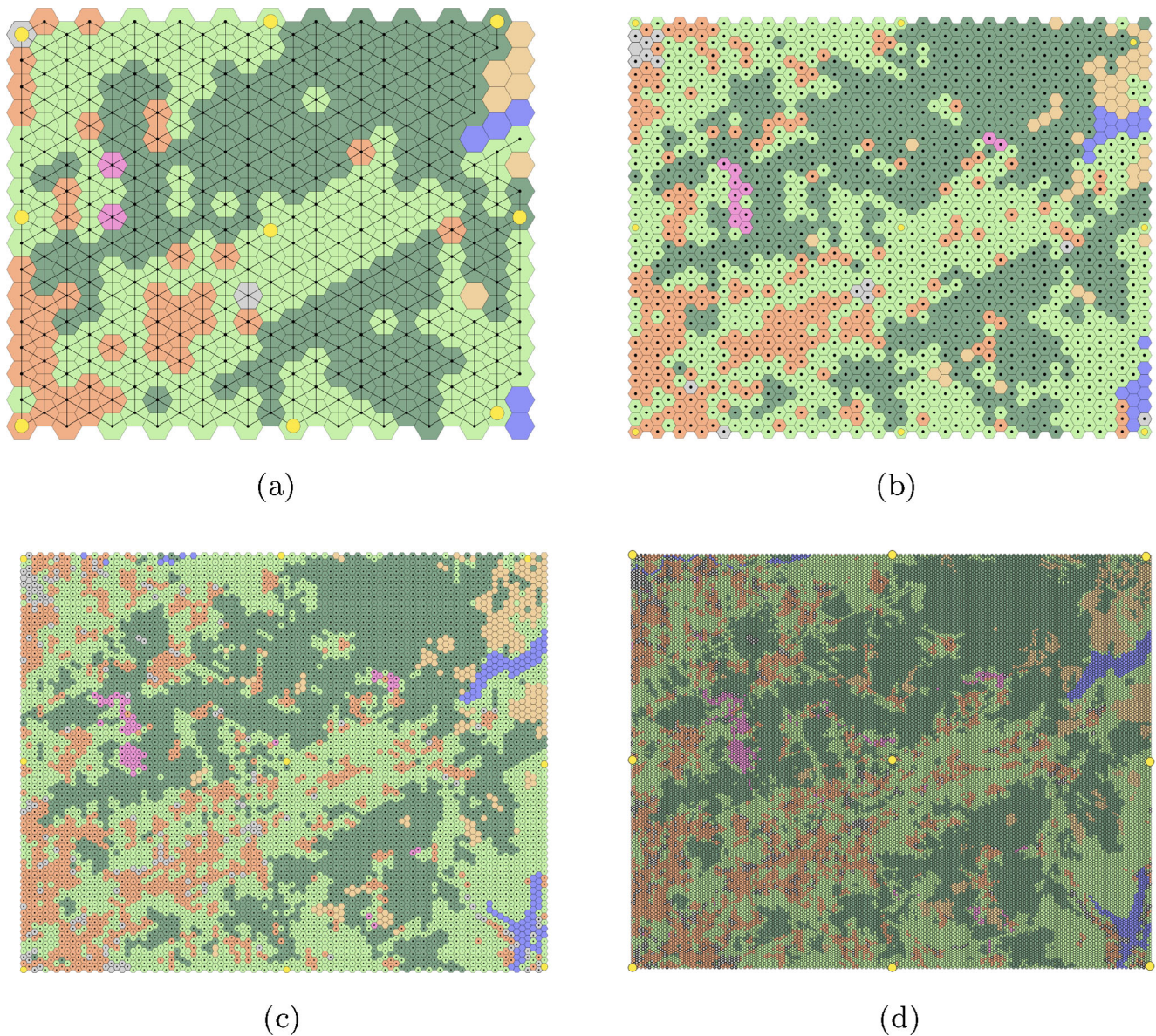


FIGURE 5 | Cells, nodes, and arcs for the four resolutions tested. Nodes colored yellow are ignition nodes in the scenarios tested—see details below. (a) Resolution of 1 000 m, (b) Resolution of 500 m (arcs are omitted for clarity), (c) Resolution of 250 m (arcs are omitted for clarity), and (d) Resolution of 100 m (arcs and nodes are omitted for clarity).

for the four resolutions used: 1 000, 500, 250, and 100 m. An arc, representing potential fire transmission, is created in each orientation between each pair of adjacent nodes. Nodes corresponding to artificialized territories only have incoming arcs—we want to keep track of the fire arrival time at those nodes but assume they do not transmit fire. This is useful for addressing other objectives, such as protecting sensible cells ([14]).

In each arc, fire transmission times between adjacent nodes are estimated based on the widely used Rothermel's fire spread model [28]. This fire spread model estimates the rate of spread (ROS) in homogeneous fuel conditions, wind, and slope. Currently, O3F provides two main ways of determining the ROS of an arc, both based on the fuel models defined for Portugal in References [29, 30]. For upslope wind, Behave 6 [31] was used to obtain the ROS for the 18 fuel models with all combinations of slope and

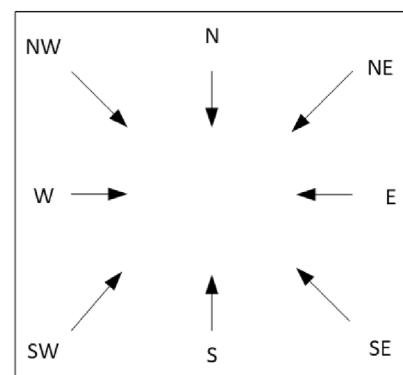


FIGURE 6 | Direction of the wind in the 8 scenarios with projected wind.

TABLE 2 | Fire spread simulation results. Resolution in meters. n/m-number of nodes/arcs of the fire network, P-number of nodes burning at alert, B-number of nodes burned from the alert until 90 min, GR-growth rate (hectares/hour) from the alert to 90 min, T-computational time (seconds).

Resolution	1 000						500						250						100						
	Scenario	n	m	P	B	GR	T	n	m	P	B	GR	T	n	m	P	B	GR	T	n	m	P	B	GR	T
	0_N_P	356	1959	5	16	635	0	1458	8218	9	60	736	0	5964	33538	24	320	1068	7	37297	210171	84	1969	1088	132
	1_NW_P	355	1956	0	0	0	0	1457	8213	2	4	29	0	5964	33546	7	46	141	2	37370	210749	0	0	0	0
	2_W_P	356	1959	1	6	289	0	1458	8218	3	19	231	0	5965	33542	9	100	328	2	37343	210541	26	603	333	157
	3_SW_P	356	1959	2	7	289	0	1458	8218	4	25	303	0	5965	33550	7	111	375	2	37358	210703	19	332	181	132
	4_S_P	356	1959	7	16	520	0	1458	8218	12	80	981	0	5964	33538	21	334	1129	3	37318	210317	82	2131	1183	129
	5_SE_P	356	1959	3	9	346	0	1458	8218	1	2	14	0	5964	33548	4	10	22	0	37344	210607	14	60	27	0
	6_E_P	356	1959	1	4	173	0	1458	8218	3	15	173	0	5965	33542	8	59	184	2	37339	210517	24	421	229	116
	7_NE_P	356	1959	2	5	173	0	1458	8218	3	4	14	0	5966	33554	1	6	18	2	37369	210745	4	32	16	112
	8_N_U	355	1953	3	39	2078	0	1455	8196	7	115	1559	0	5929	33322	18	583	2039	2	37096	209071	49	3578	2037	173
	9_NW_U	355	1956	0	0	0	0	1453	8205	1	64	909	0	5957	33520	4	249	884	3	37370	210749	0	0	0	0
	10_W_U	355	1951	4	39	2021	0	1447	8144	10	143	1920	0	5913	33206	23	588	2039	3	37074	208929	55	3340	1897	241
	11_SW_U	356	1959	1	10	520	0	1456	8208	3	65	895	0	5959	33534	2	242	866	2	37328	210606	4	1277	735	267
	12_S_U	355	1953	3	36	1905	0	1454	8188	8	146	1992	0	5932	33340	16	552	1934	2	37113	209188	45	3434	1957	281
	13_SE_U	355	1953	3	31	1617	0	1456	8217	0	0	0	0	5962	33541	3	12	32	0	37336	210582	10	64	31	0
	14_E_U	356	1959	2	21	1097	0	1456	8204	5	72	967	0	5943	33422	14	261	891	2	37200	209757	27	1758	999	190
	15_NE_U	356	1959	1	4	173	0	1456	8212	2	13	159	0	5958	33528	2	35	119	2	37364	210729	2	149	85	155
	16_C_U	356	1959	6	82	4388	0	1450	8146	16	310	4244	0	5916	33198	34	1055	3684	2	36962	208242	80	6484	3697	220

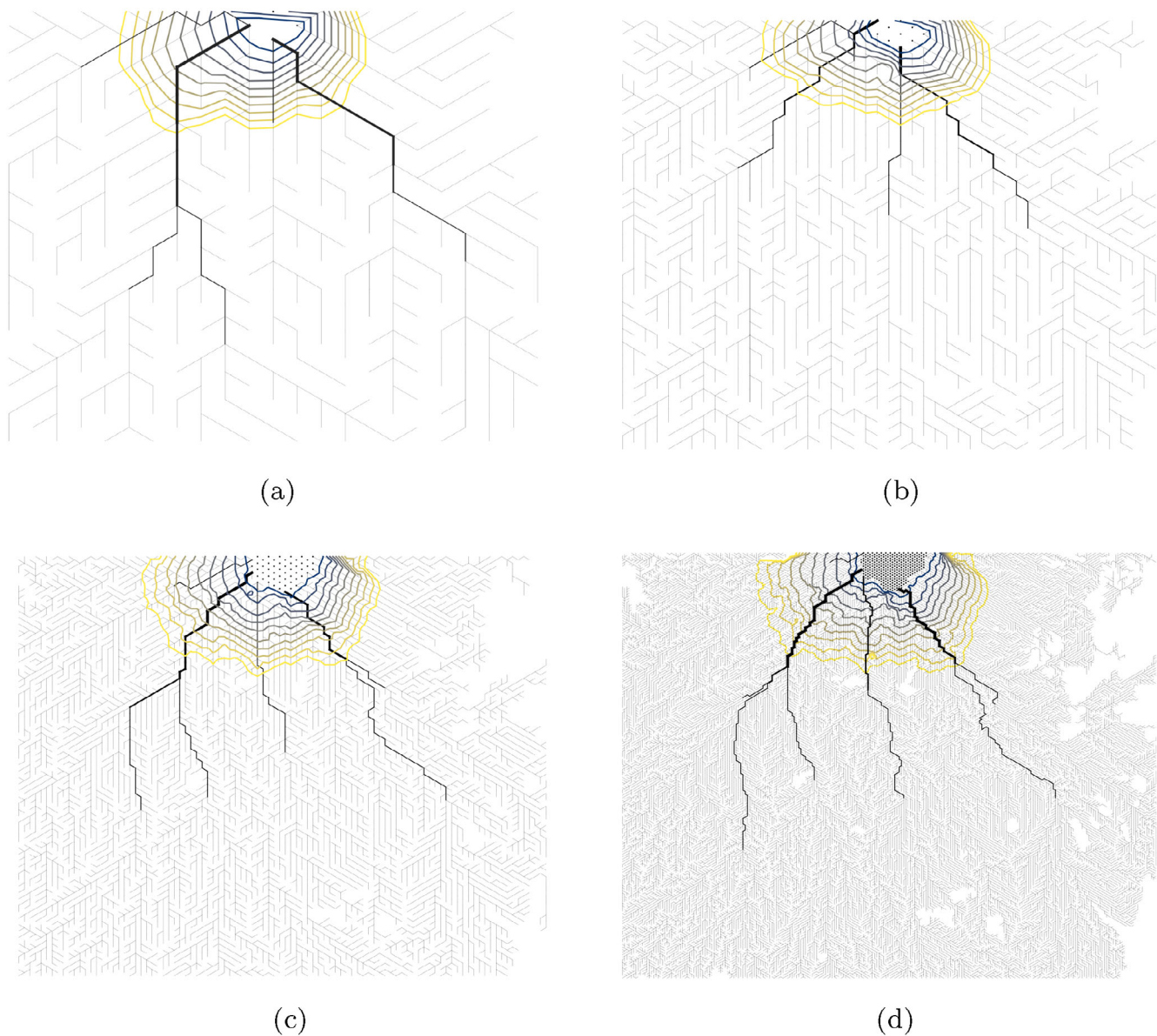


FIGURE 7 | Scenario north ignition and upslope wind. Ten isochronous curves are represented with instants from 0 to 90 min. Dots correspond to burned or burning nodes at the alert instant. Fire paths are also represented. Five category widths for the arcs are considered according to the number of paths that include each arc. (a) Resolution of 1 000 m, (b) Resolution of 500 m, (c) Resolution of 250 m, and (d) Resolution of 100 m.

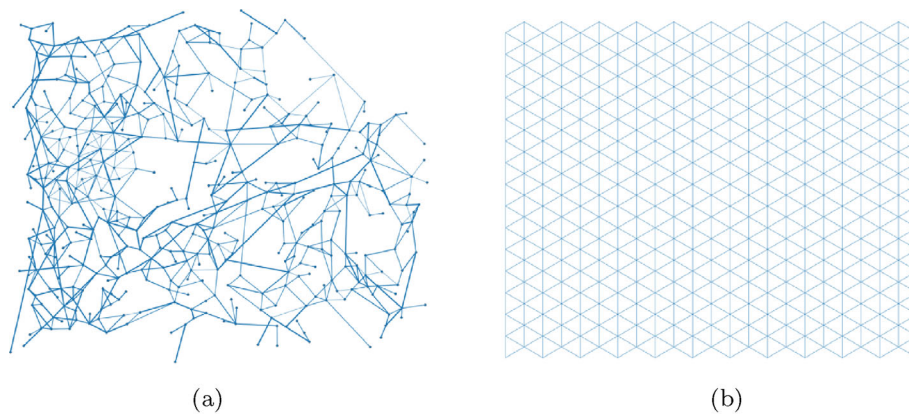


FIGURE 8 | Networks for the classes of resources. (a) Network for terrestrial resources-roads with parameter contraction 500 m and (b) Network for air resources (500 m resolution).

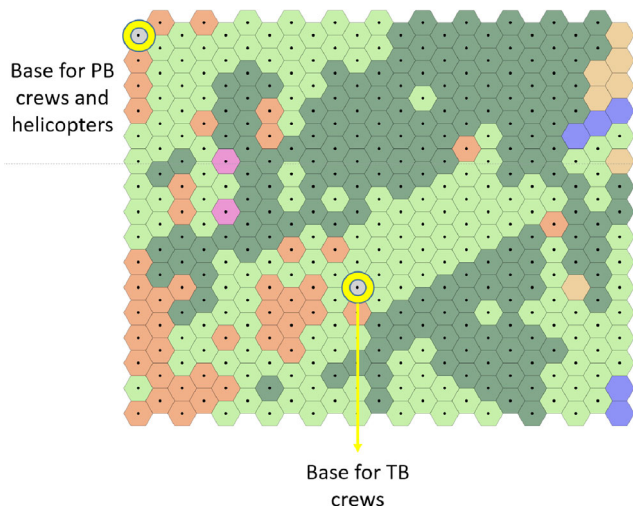


FIGURE 9 | Bases for the resources.

TABLE 3 | Characterization of the tested combinations of resources.

Combination	Type(s)		Availability		Helicopter(s)	
	single	multiple	high	low	yes	no
(10,0,0)	X		X		X	
(0,10,0)	X		X			X
(0,0,10)	X		X			X
(10,10,10)		X	X		X	
(1,3,3)		X		X	X	
(0,3,3)		X		X		X

upslope wind intensity. For the wind in a direction different from the upslope direction, Behave 6 was used to obtain the no-wind, no-slope ROS (i.e., R_0) and the projection method described in Reference [32] was used to obtain the ROS in the desired direction (i.e., the arc direction).

To determine the ROS of each arc, each land use is associated with a set of possible fuel models. For example, the eucalyptus forest (land use level 4 code 5.1.1.4) is associated with fuel models 211, 223, 224, 225, and 226. In the present work, the average ROS is considered. The ROS estimate between two nodes with different fuels is obtained by the average of both ROS. Naturally, higher resolutions lead to better ROS approximations. Characterizing fuels and estimating ROS are research topics on their own, and their treatment is outside the scope of this work. In any case, it is worth mentioning that the methods described in this paper use ROS only as a parameter and, therefore, virtually any method for obtaining the fire ROS can be used without any change in the model.

6.2 | Scenarios and Fire Spread Simulation

To experiment with the conceived approaches 17 scenarios were considered. Each scenario is defined by an ignition node and a type of wind (P-projected or U-upslope, see previous section). The wind speed is set to 30 km/h. Although more than one ignition could be considered, it was assumed that a single ignition in each

scenario occurs. The fire alert is given 30 min after the ignition. The evaluation instant is 90 min after the fire alert occurs (i.e., 120 min after the ignition) aligned with the recommendations of the National Emergency and Civil Protection Command for Fighting Rural Fires 2023 that state that “the initial attack is an organized and integrated with a duration up to 90 min after the alert” [6]. As ignition nodes, the node closest to each cardinal point of the landscape (N, NW, W, SW, S, SE, E, NE) were considered. In the scenarios with projected wind, the wind is in the direction of the center of the landscape as shown in Figure 6. For upslope wind, a scenario with an ignition at the center is also considered. Ignition nodes are colored yellow in Figure 5.

We first consider the simulation of fire spread with no resources obtained by a shortest path tree algorithm. Table 2 presents the main results for the fire spread simulation of the 17 scenarios and the four resolutions. We use the growth rate (GR), defined as the burned area per unit of time, as the measure of fire spread, allowing comparisons for the same scenario at different resolutions and with actual fires.

Fire spread is significantly different for different scenarios. In some scenarios, fire doesn't spread, while in others fire spreads with a very high growth rate. In the scenario with the highest growth rates (16_C_U), the values for the different resolutions are about half the growth rate of the 2017 extreme fire of Pedrogão Grande, Portugal [33].

Scenarios where fire does not spread, or have a low growth rate, have ignitions in nodes corresponding to artificialized cells (that do not transmit fire, as mentioned before) or in cells belonging to a small component (i.e., few cells surrounded by unburnable cells).

Scenarios with upslope wind tend to have a much higher GR than the ones defined with projected wind. For the 100 m resolution and the most difficult instances, the computational times are less than 5 min. For the other resolutions, the fire spread simulation takes a few seconds at most.

While the differences in terms of GR may be significant when comparing the GR of resolutions 1 000 to 500 and 500 to 250, the difference between 250 and 100 resolution is smaller. Considering the scenarios with an average GR of at least 500 ha/h (eight scenarios), the maximum relative difference between resolutions 1 000 and 500, 500 and 250, and 250 and 100 are, respectively, 89%, 45%, and 15%. The average relative difference is 28%, 16%, and 5%.

As an example, Figure 7 presents the simulation of fire spread for scenario 8 (north ignition, upslope wind) at four resolutions.

6.3 | MIP and Resources

As described before, we consider two classes of resources: Terrestrial and aerial resources. Each class has a network that can be seen as a layer when integrated with fire spread. The terrestrial network is based on the roads taken from OpenStreetMap [34] and transformed to a graph which is further processed to avoid nodes too close to each other (a contraction parameter controls which nodes are contracted). Roads have different types (from

TABLE 4 | Results for MIP with resolution 1 000 m. n and m-number of nodes and arcs of the fire network after preprocessing, UB/LB upper bound/lower bound on the number of nodes burned from the alert until 90 min provided by the solver, G-relative gap at the end of the optimization in percentage, $G = 100 * (UB - LB)/UB$, T-computational time (seconds)-a time limit of 1 800 s was set.

Resources			(10,0,0)				(0,10,0)				(0,0,10)				(10,10,10)				(1,3,3)				(0,3,3)			
Scenario	n	m	UB	LB	G	T	UB	LB	G	T	UB	LB	G	T	UB	LB	G	T	UB	LB	G	T	UB	LB	G	T
0_N_P	16	30	9	9	0	0	8	7	13	1801	7	7	0	1	7	7	0	2	7	7	0	0	7	7	0	0
1_NW_P	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
2_W_P	6	18	3	3	0	0	3	3	0	14	3	3	0	2	3	3	0	20	3	3	0	2	3	3	0	1
3_SW_P	7	12	4	4	0	0	4	3	25	1802	3	3	0	1	3	3	0	2	3	3	0	1	3	3	0	1
4_S_P	16	30	12	12	0	0	16	8	50	1802	10	9	10	1801	10	8	20	1802	10	8	20	1802	10	8	20	1803
5_SE_P	9	26	7	7	0	0	8	8	0	2	8	8	0	1	7	7	0	2	7	7	0	1	8	8	0	1
6_E_P	4	8	3	3	0	0	3	3	0	5	3	3	0	1	3	3	0	2	3	3	0	0	3	3	0	0
7_NE_P	5	8	4	4	0	0	5	4	20	1802	4	4	0	1	4	4	0	2	4	4	0	1	4	4	0	1
8_N_U	39	182	16	16	0	0	23	18	22	1802	26	11	58	1801	13	11	15	1801	19	11	42	1801	23	11	52	1801
9_NW_U	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
10_W_U	39	182	21	21	0	0	23	19	17	1801	26	19	27	1801	21	19	10	1801	21	19	10	1801	23	19	17	1801
11_SW_U	10	36	3	3	0	0	6	6	0	36	6	6	0	1	3	3	0	1	3	3	0	0	6	6	0	1
12_S_U	36	160	21	21	0	0	29	10	66	1801	22	10	55	1804	19	10	47	1802	19	10	47	1802	22	10	55	1801
13_SE_U	31	138	16	16	0	0	25	16	36	1800	25	16	36	1801	16	15	6	1801	22	15	32	1801	25	16	36	1801
14_E_U	21	80	11	11	0	0	14	12	14	1802	13	13	0	132	11	11	0	37	11	10	9	1801	13	12	8	1801
15_NE_U	4	8	3	3	0	0	4	4	0	0	4	4	0	0	3	3	0	0	3	3	0	0	4	4	0	0
16_C_U	82	414	35	35	0	0	48	30	38	1801	30	30	0	5	28	28	0	12	32	32	0	127	46	43	7	1802

TABLE 5 | Results for MIP with resolution 500 m. n and m-number of nodes and arcs of the fire network after preprocessing, UB/LB upper bound/lower bound on the number of nodes burned from the alert until 90 min provided by the solver, G-relative gap at the end of the optimization in percentage, $G = 100 * (UB - LB)/UB$, T-computational time (seconds)-a time limit of 1800 s was set.

Resources			(10,0,0)					(0,0,10)					(10,10,10)					(1,3,3)					(0,3,3)				
Scenario	n	m	UB	LB	G	T	UB	LB	G	T	UB	LB	G	T	UB	LB	G	T	UB	LB	G	T	UB	LB	G	T	
0_N_P	60	218	22	22	0	0	34	21	38	1801	23	21	9	1801	19	17	11	1801	20	18	10	1801	23	21	9	1801	
1_NW_P	4	5	4	4	0	0	4	4	0	0	4	4	0	0	4	4	0	0	4	4	0	0	4	4	0	0	
2_W_P	19	80	10	10	0	0	14	6	57	1801	6	6	0	13	6	6	0	30	6	6	0	2	6	6	0	11	
3_SW_P	25	80	13	13	0	0	15	9	40	1802	11	9	18	1801	11	9	18	1801	11	9	18	1801	11	9	18	1801	
4_S_P	80	342	50	50	0	0	59	35	41	1802	52	35	33	1802	47	29	38	1801	50	33	34	1801	52	35	33	1801	
5_SE_P	2	1	2	2	0	0	2	2	0	0	2	2	0	0	2	2	0	0	2	2	0	0	2	2	0	0	
6_E_P	15	60	9	9	0	0	12	6	50	1802	12	6	50	1802	9	6	33	1801	9	6	33	1801	12	6	50	1802	
7_NE_P	4	6	4	4	0	0	4	4	0	0	4	4	0	0	4	4	0	0	4	4	0	0	4	4	0	0	
8_N_U	115	592	33	33	0	0	63	47	25	1802	63	47	25	1801	42	33	21	1802	48	39	19	1809	60	47	22	1801	
9_NW_U	64	309	5	5	0	0	5	5	0	2	5	5	0	20	5	5	0	5	5	5	0	1	5	5	0	1	
10_W_U	143	750	43	43	0	0	62	36	42	1801	58	36	38	1802	42	35	17	1802	52	35	33	1801	60	40	33	1802	
11_SW_U	65	320	25	25	0	0	37	8	78	1801	16	8	50	1802	14	8	43	1802	14	8	43	1802	16	8	50	1802	
12_S_U	146	734	58	58	0	0	105	72	31	1802	91	72	21	1802	58	48	17	1803	85	59	31	1807	98	75	23	1801	
13_SE_U	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
14_E_U	72	350	35	35	0	0	51	42	18	1801	44	42	5	1801	34	27	21	1801	37	31	16	1802	44	38	14	1802	
15_NE_U	13	42	7	7	0	0	13	13	0	0	13	13	0	0	7	7	0	1	7	7	0	0	13	13	0	0	
16_C_U	310	1658	96	96	0	1	270	97	64	1801	158	97	39	1801	84	76	10	1801	197	123	38	1801	217	151	30	1802	

TABLE 6 | Number of burned nodes: MIP versus PRH for resolution 1 000 m. Differences signaled in bold.

Resources			(10,0,0)		(0,10,0)		(0,0,10)		(10,10,10)		(1,3,3)		(0,3,3)	
Scenario	n	m	MIP	PRH	MIP	PRH	MIP	PRH	MIP	PRH	MIP	PRH	MIP	PRH
0_N_P	16	30	9	9	8	8	7	7	7	7	7	7	7	7
1_NW_P	0	0	0	0	0	0	0	0	0	0	0	0	0	0
2_W_P	6	18	3	3	3	3	3	3	3	3	3	3	3	3
3_SW_P	7	12	4	4	4	4	3	3	3	3	3	3	3	3
4_S_P	16	30	12	12	16	15	10	11	10	11	10	11	10	11
5_SE_P	9	26	7	7	8	8	8	8	7	7	7	7	8	8
6_E_P	4	8	3	3	3	3	3	3	3	3	3	3	3	3
7_NE_P	5	8	4	4	5	5	4	4	4	4	4	4	4	4
8_N_U	39	182	16	16	23	23	26	26	13	13	19	18	23	23
9_NW_U	0	0	0	0	0	0	0	0	0	0	0	0	0	0
10_W_U	39	182	21	21	23	23	26	26	21	21	21	21	23	23
11_SW_U	10	36	3	3	6	6	6	6	3	3	3	3	6	6
12_S_U	36	160	21	21	29	28	22	22	19	19	19	19	22	22
13_SE_U	31	138	16	16	25	25	25	25	16	16	22	22	25	25
14_E_U	21	80	11	11	14	14	13	13	11	11	11	11	13	13
15_NE_U	4	8	3	3	4	4	4	4	3	3	3	3	4	4
16_C_U	82	414	35	35	48	48	30	30	28	28	32	32	46	44

TABLE 7 | Number of burned nodes: MIP versus PRH for resolution 500 m.

Resources			(10,0,0)		(0,10,0)		(0,0,10)		(10,10,10)		(1,3,3)		(0,3,3)	
Scenario	n	m	MIP	PRH	MIP	PRH	MIP	PRH	MIP	PRH	MIP	PRH	MIP	PRH
0_N_P	60	218	22	22	34	34	23	23	19	19	20	20	23	23
1_NW_P	4	5	4	4	4	4	4	4	4	4	4	4	4	4
2_W_P	19	80	10	10	14	12	6	6	6	6	6	6	6	6
3_SW_P	25	80	13	13	15	15	11	11	11	11	11	11	11	11
4_S_P	80	342	50	50	59	59	52	52	47	47	50	50	52	52
5_SE_P	2	1	2	2	2	2	2	2	2	2	2	2	2	2
6_E_P	15	60	9	9	12	12	12	12	9	9	9	9	12	12
7_NE_P	4	6	4	4	4	4	4	4	4	4	4	4	4	4
8_N_U	115	592	33	33	63	63	63	63	42	33	48	48	60	60
9_NW_U	64	309	5	5	5	5	5	5	5	5	5	5	5	5
10_W_U	143	750	43	43	62	58	58	53	42	42	52	49	60	55
11_SW_U	65	320	25	25	37	37	16	16	14	14	14	14	16	16
12_S_U	146	734	58	58	105	107	91	91	58	58	85	79	98	95
13_SE_U	0	0	0	0	0	0	0	0	0	0	0	0	0	0
14_E_U	72	350	35	35	51	51	44	44	34	34	37	37	44	44
15_NE_U	13	42	7	7	13	13	13	13	7	7	7	7	13	13
16_C_U	310	1 658	96	96	270	220	158	152	84	84	197	169	217	205

TABLE 8 | PRH results (resolution 250 m) for sets of resources with only type available. n/m-number of nodes/arcs of the fire network after preprocessing, GRS-a growth rate of the simulation (no resources available) in ha/h, R-resources used, B-number of burned nodes from the alert until 90 min, C-the cost of the used resources, GR-growth rate, Δ GRS-a variation of the GR with respect to GRS, Δ GRP-a variation of the GR with respect to the positioning without taking into account that fire cuts the route (only positioning, no routing), O-number of cycles of the PRH, T-computational time (seconds). * Minimum, maximum, and average values are for scenarios with a GRS higher than 500 ha/h.

Resources			None		(10,0,0)							(0,10,0)							(0,0,10)								
Scenario	n	m	GRS	R	B	C	GR	ΔGRS	ΔGRP	O	T	R	B	C	GR	ΔGRS	ΔGRP	O	T	R	B	C	GR	ΔGRS	ΔGRP	O	T
0_N_P	320	1 626	1 068	(10, 0, 0)	128	100	375	-693	0	1	3	(0, 7, 0)	169	7	523	-545	83	5	1	(0, 0, 5)	166	5	512	-556	10	2	0
1_NW_P	46	154	141	(1, 0, 0)	20	10	47	-94	0	1	0	(0, 1, 0)	19	1	43	-98	0	1	0	(0, 0, 1)	22	1	54	-87	0	1	0
2_W_P	100	514	328	(3, 0, 0)	36	30	97	-231	0	1	0	(0, 3, 0)	43	3	123	-205	0	1	0	(0, 0, 3)	41	3	115	-213	0	1	0
3_SW_P	111	487	375	(4, 0, 0)	41	40	123	-252	0	1	0	(0, 4, 0)	71	4	231	-144	79	4	0	(0, 0, 3)	39	3	115	-260	0	1	0
4_S_P	334	1 688	1 129	(10, 0, 0)	115	100	339	-790	0	1	1	(0, 5, 0)	168	5	530	-599	0	1	0	(0, 0, 5)	155	5	484	-645	4	2	0
5_SE_P	10	21	22	(0, 0, 0)	10	0	22	0	0	1	0	(0, 0, 0)	10	0	22	0	0	1	0	(0, 0, 0)	10	0	22	0	0	1	0
6_E_P	59	284	184	(4, 0, 0)	33	40	90	-94	0	1	0	(0, 4, 0)	45	4	134	-50	0	1	0	(0, 0, 4)	45	4	134	-50	0	1	0
7_NE_P	6	12	18	(1, 0, 0)	2	10	4	-14	0	1	0	(0, 0, 0)	6	0	18	0	0	1	0	(0, 0, 0)	6	0	18	0	0	1	0
8_N_U	583	3 203	2 039	(7, 0, 0)	127	70	393	-1 646	0	1	3	(0, 8, 0)	306	8	1 039	-1 000	0	1	1	(0, 0, 8)	306	8	1 039	-1 000	0	1	1
9_NW_U	249	1 218	884	(2, 0, 0)	25	20	76	-808	0	1	1	(0, 2, 0)	12	2	29	-855	0	1	0	(0, 0, 2)	57	2	191	-693	0	1	0
10_W_U	588	3 229	2 039	(7, 0, 0)	134	70	401	-1 638	0	1	2	(0, 9, 0)	185	9	585	-1 454	0	1	1	(0, 0, 10)	182	10	574	-1 465	0	1	1
11_SW_U	242	1 250	866	(4, 0, 0)	54	40	188	-678	0	1	1	(0, 4, 0)	70	4	245	-621	14	2	0	(0, 0, 4)	47	4	162	-704	0	1	0
12_S_U	552	2 972	1 934	(9, 0, 0)	165	90	538	-1 396	0	1	2	(0, 10, 0)	332	10	1 140	-794	0	1	1	(0, 0, 10)	256	10	866	-1 068	43	2	1
13_SE_U	12	27	32	(0, 0, 0)	12	0	32	0	0	1	0	(0, 0, 0)	12	0	32	0	0	1	0	(0, 0, 0)	12	0	32	0	0	1	0
14_E_U	261	1 352	891	(8, 0, 0)	114	80	361	-530	0	1	1	(0, 7, 0)	189	7	631	-260	0	1	0	(0, 0, 7)	180	7	599	-292	0	1	0
15_NE_U	35	136	119	(2, 0, 0)	12	20	36	-83	0	1	0	(0, 0, 0)	35	0	119	0	0	1	0	(0, 0, 0)	35	0	119	0	0	1	0
16_C_U	1 055	5 858	3 684	(10, 0, 0)	257	100	805	-2 879	0	1	7	(0, 10, 0)	662	10	2 266	-1 418	281	10	13	(0, 0, 10)	458	10	1 530	-2 154	307	3	8
\$min*^\$	242	1 218	866		25	20	76	-2 879	0	1	1		12	2	29	-1 454	0	1	0		47	2	162	-2 154	0	1	0
\$max*^\$	1 055	5 858	3 684		257	100	805	-530	0	1	7		662	10	2 266	-260	281	10	13		458	10	1 530	-292	307	3	8
\$avg*^\$	465	2 488	1 615		124	74	386	-1 229	0	1	2		233	7	776	-838	42	3	2		201	7	662	-953	40	2	1

TABLE 9 | PRH results (resolution 250 m) for sets of resources with more than one type available. n/m-number of nodes/arcs of the fire network after preprocessing, GRS-a growth rate of the simulation (no resources available) in ha/h, R-resources used, B-number of burned nodes from the alert until 90 min, C-the cost of the used resources, GR-growth rate, ΔGRS-a variation of the GR with respect to GRS, ΔGRP-a variation of the GR with respect to the positioning without taking into account that fire cuts the route (only positioning, no routing), O-number of cycles of the PRH, T-computational time (seconds).
* Minimum, maximum, and average values are for scenarios with a GRS higher than 500 ha/h.

Resources			None		(10,10,10)										(1,3,3)					(0,3,3)							
Scenario	n	m	GRS	R	B	C	GR	AGRS	ΔGRP	O	T	R	B	C	GR	AGRS	ΔGRP	O	T	R	B	C	GR	AGRS	ΔGRP	O	T
0_N_P	320	1626	1068	(8, 2, 1)	118	83	339	-729	0	4	7	(1, 3, 3)	149	16	451	-617	54	6	19	(0, 3, 3)	163	6	502	-566	76	5	2
1_NW_P	46	154	141	(0, 1, 0)	19	1	43	-98	0	1	0	(0, 1, 0)	19	1	43	-98	0	1	0	(0, 1, 0)	19	1	43	-98	0	1	0
2_W_P	100	514	328	(2, 1, 1)	34	22	90	-238	0	1	0	(1, 1, 1)	36	12	97	-231	0	1	0	(0, 1, 2)	41	3	115	-213	0	1	0
3_SW_P	111	487	375	(1, 0, 2)	36	12	105	-270	0	1	0	(1, 0, 2)	36	12	105	-270	0	1	0	(0, 0, 3)	39	3	115	-260	0	1	0
4_S_P	334	1688	1129	(7, 0, 3)	113	73	332	-797	0	1	2	(1, 2, 3)	140	15	429	-700	3	3	6	(0, 2, 3)	155	5	484	-645	4	3	1
5_SE_P	10	21	22	(0, 0, 0)	10	0	22	0	0	1	0	(0, 0, 0)	10	0	22	0	0	1	0	(0, 0, 0)	10	0	22	0	0	1	0
6_E_P	59	284	184	(3, 0, 1)	33	31	90	-94	0	1	0	(1, 0, 2)	38	12	108	-76	0	1	0	(0, 1, 3)	45	4	134	-50	0	1	0
7_NE_P	6	12	18	(1, 0, 0)	2	10	4	-14	0	1	0	(1, 0, 0)	2	10	4	-14	0	1	0	(0, 0, 0)	6	0	18	0	0	1	0
8_N_U	583	3203	2039	(7, 0, 0)	127	70	393	-1646	0	1	3	(1, 3, 3)	242	16	808	-1231	0	1	7	(0, 3, 3)	317	6	1079	-960	0	1	2
9_NW_U	249	1218	884	(0, 2, 0)	12	2	29	-855	0	1	1	(0, 2, 0)	12	2	29	-855	0	1	2	(0, 2, 0)	12	2	29	-855	0	1	0
10_W_U	588	3229	2039	(5, 0, 2)	130	52	386	-1653	0	1	5	(1, 3, 3)	153	16	469	-1570	0	1	10	(0, 3, 3)	199	6	635	-1404	0	1	8
11_SW_U	242	1250	866	(1, 0, 2)	39	12	134	-732	0	1	1	(1, 0, 2)	39	12	134	-732	0	1	1	(0, 1, 3)	47	4	162	-704	0	1	0
12_S_U	552	2972	1934	(6, 0, 3)	154	63	498	-1436	22	2	6	(1, 3, 3)	208	16	693	-1241	51	2	14	(0, 3, 3)	296	6	1010	-924	18	2	5
13_SE_U	12	27	32	(0, 0, 0)	12	0	32	0	0	1	0	(0, 0, 0)	12	0	32	0	0	1	0	(0, 0, 0)	12	0	32	0	0	1	0
14_E_U	261	1352	891	(6, 0, 1)	111	61	350	-541	0	1	1	(1, 2, 3)	142	15	462	-429	0	1	1	(0, 3, 3)	181	6	603	-288	0	1	1
15_NE_U	35	136	119	(2, 0, 0)	12	20	36	-83	0	1	0	(1, 0, 0)	15	10	47	-72	0	1	0	(0, 0, 0)	35	0	119	0	0	1	0
16_C_U	1055	5858	3684	(9, 0, 6)	215	96	653	-3031	0	1	6	(1, 3, 3)	427	16	1418	-2266	105	3	407	(0, 3, 3)	593	6	2017	-1667	357	4	76
\$min*%\$	242	1218	866		12	2	29	-3031	0	1	1		12	2	29	-2266	0	1	1		12	2	29	-1667	0	1	0
\$max*%\$	1055	5858	3684		215	96	653	-541	22	4	7		427	16	1418	-429	105	6	407		593	6	2017	-288	357	5	76
\$avg*%\$	465	2488	1615		113	57	346	-1269	2	1	4		168	14	544	-1071	24	2	52		218	5	725	-890	51	2	11

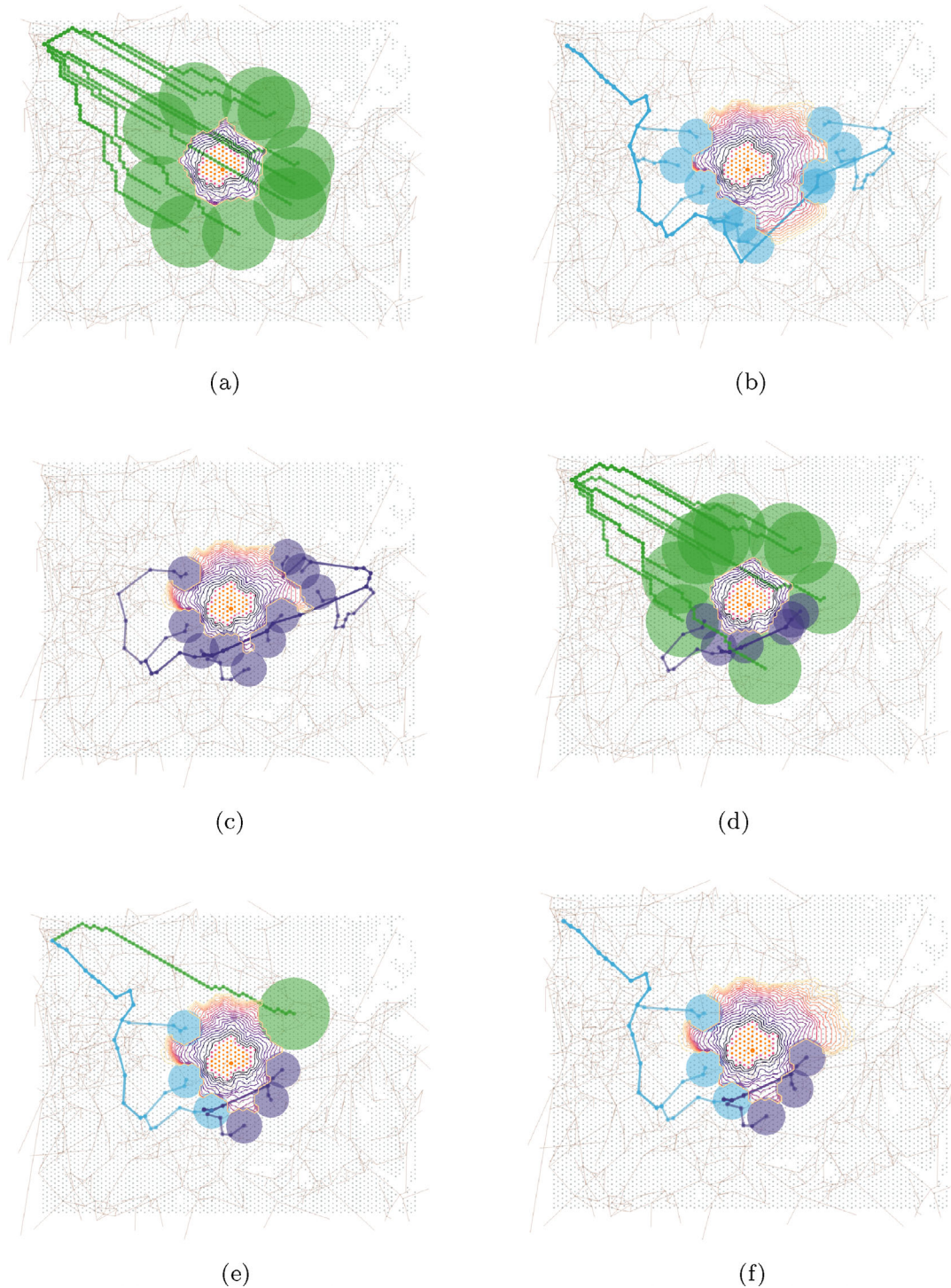


FIGURE 10 | Final solution of the PRH for the most difficult scenario (upslope wind and ignition at the center). Grey nodes correspond to fire nodes, grey lines to roads. Green represents the attack area and the route of aerial resources, while blue/violet represents the attack area of terrestrial resource units based at PB/TB. Burned and burning nodes at the alert are represented in the center. Isochronous curves are related to fire spread until 90 min.

major roads to paths unsuitable for cars), in which terrestrial resources have different speeds. Aerial resources move and attack fire on a grid. Examples of terrestrial and aerial resource networks are given in Figure 8.

It is important to note that two resources may share the same class (e.g., terrestrial resources share the same road network) but

have different types: Their base may be different or they may have different speeds. In these experiments, we considered fire-fighting crews (class terrestrial) and helicopters (class aerial). For the former, we considered two types according to their base location: Ponte da Barca (PB) and Terras de Bouro (TB), as shown in Figure 9. The base for helicopters (H) is Ponte da Barca. Each resource has a given number of available units.

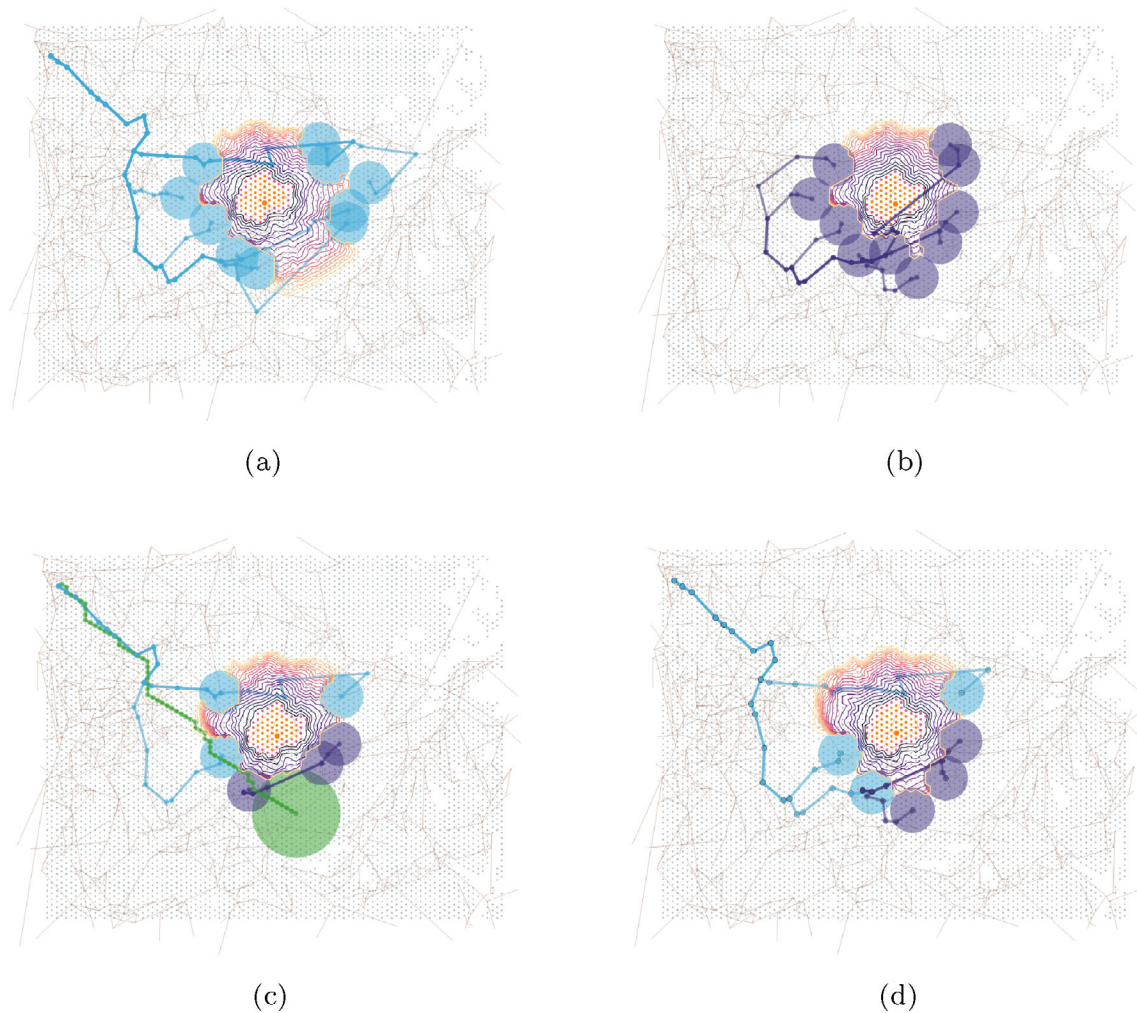


FIGURE 11 | First solution of the PRH for the most difficult scenario (upslope wind and ignition at the center), when different from the final solution provided by the PRH. Grey nodes correspond to fire nodes, grey lines to roads. Green represents the attack area and the route of aerial resources, while blue/violet represents the attack area of terrestrial resource units based at PB/TB. Burned and burning nodes at the alert are represented in the center. Isochronous curves are related to fire spread until 90 min.

For the crews, we set the preparation time to 5 min, the suppression radius to 1 000 m, and the unsafety radius to 500 m. For the helicopters, the preparation time was set to 15 min and the suppression radius to 2 000 m. The cost of one helicopter is 10 times the cost of one crew.

We represent the available resources as (H, PB, and TB), where H is the number of helicopters, and PB and TB are the number of crews with bases in PB and TB, respectively. We tested six diverse combinations of the number of units of the three types of resources: Single type versus multiple types, high availability versus low availability, and helicopter versus no helicopter. The characterization of each combination is provided in Table 3.

Tables 4 and 5 are relative to the results of the MIP model solved directly for resolutions of 1 000 and 500 m, respectively. For each scenario, indicated in the left column, results for each combination of resources, indicated in the first row, are presented. The meaning of the values shown in the tables is described in their caption.

A first observation is that the preprocessing procedure described at the end of Section 4 is effective in reducing the size of the fire network, as can be seen by comparing the number of nodes and arcs shown in Tables 4 and 5 with those in the columns corresponding to 1 000 and 500 m of resolution in Table 2. Several scenarios become trivial given their reduced number of nodes and arcs. Even so, the MIP approach reaches the limit of 1 800 s in many scenarios with high gaps.

We detail the results of the positioning and routing heuristic (PRH) in the next section. For now we compare the results obtained by solving the MIP model directly and the PRH. The comparison in terms of the number of burned nodes is presented in Tables 6 and 7. The PRH provided a better solution than direct MIP in 16 scenarios/sets of resources and the worst solution in five.

The computational time of the PRH was 0 s for all scenarios and sets of resources except six combinations that took 1 s and scenario 16_C_U with resolution 500 that took 3, 4, 2, 113, and 27 s for the available resources, excluding (10,0,0), in the same

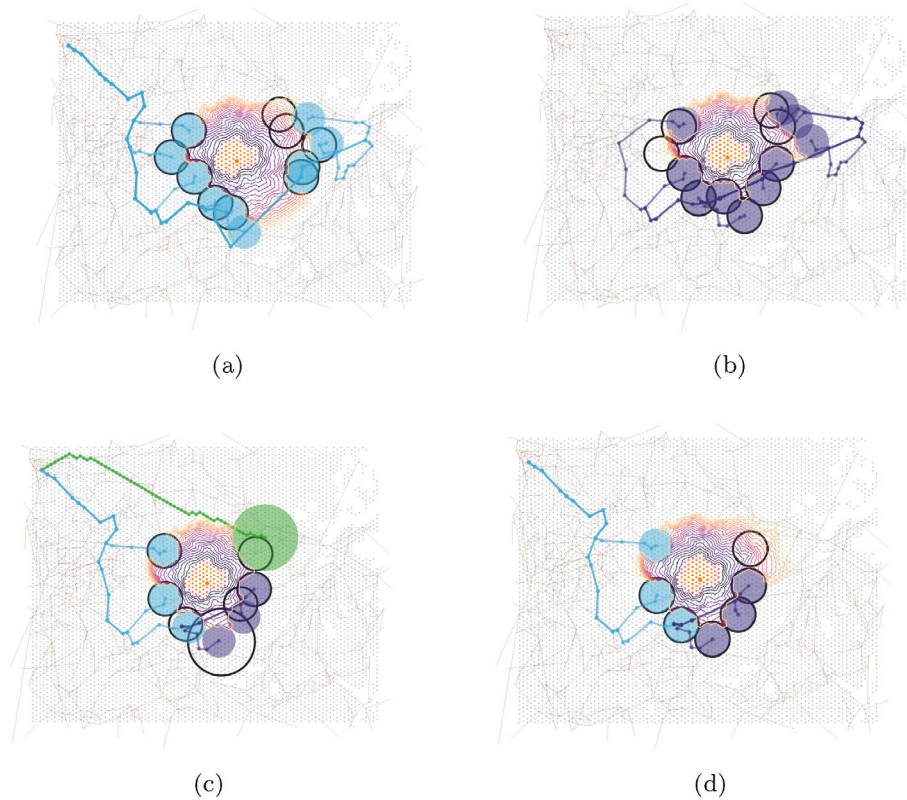


FIGURE 12 | Comparison of solutions with positioning only (first PRH solution) and PRH for the most difficult scenario (upslope wind and ignition at the center). All elements from the PRH solution except the circles with bold borders representing the position of a unit in the first solution.

order as in the tables. Scenario 16_C_U with resources (1,3,3), the one that took considerably more time, implied 14 iterations, that is, 14 MIP optimizations/PSP solved, ending with a solution significantly better (28 fewer burned nodes) than applying MIP directly.

The difference in the number of burned nodes is not so significant (except in the most difficult cases, resolution 500, 16_C_U) showing that MIP reaches good quality solutions but the lower bound it provides is poor. In the next section, we provide the results for a resolution of 250 m, where, taking into account the results of this subsection and the computational requirements for solving the MIP model directly, only the PRH was tested.

6.4 | Positioning and Routing Heuristic

In Tables 8 and 9 we present the results of the PRH for a resolution of 250 m and the same six combinations of resources. The network for aerial resources is a grid with a resolution of 250 m. The network for the terrestrial resources has 719 nodes and 2022 arcs (contraction parameter 250 m). With the exception of two scenario/resource pairs, computational times are less than 15 s, but very often close to zero. Scenario 16_C_U takes considerably more time when few resources are available (approximately 7 min and 1 min for (1,3,3) and (0,3,3) sets of available resources).

The GR reduction is significant when compared to the absence of resources. Considering only scenarios where the GR is higher

than 500 ha/h, the average reduction in GR is 78% when the number of available resources of the three types is high (10,10,10) and greater than 75% when 10 aerial resources are available (10,0,0). When many aerial resources are available, terrestrial resources make a small contribution to reducing the GR. For 10 terrestrial resources, the reduction is around 60% with a better value for the resources with a base location in TB (0,0,10) than in PB (0,10,0) given its more central position. When only three terrestrial resources are available, the use of an aerial resource allows the improvement of GR reduction from 59% to 68%.

As expected, the burned area increases in most of the scenarios (in particular the more difficult ones) when routing (PRH) is considered when comparing to only positioning (i.e., the solution of the first PSP) as can be seen by the variation presented in column ΔGRP . This justifies the relevance of addressing routing in a feasible way. Neglecting the influence of fire on the routes of terrestrial resources leads to better GR but infeasible routes in practice. In many scenarios/combinations of resources, several iterations of the PRH are needed to reach a feasible routing of the terrestrial resources.

In Figure 10, the solutions obtained with the PRH in the most difficult scenario are presented for the six combinations of resources. It can be seen that, in scenarios with 10 aerial resources available, the fire is contained.

Figure 11 presents the optimal solution of the *first* PSP, that is, the solution does not take into account the feasibility of the routes in terms of fire. Combinations of resources (10,0,0) and (10,10,10)

are not shown because the solutions are the same as those provided by the PRH. For the other combinations, it can be seen that at least one resource route crosses the fire. In Figure 12, the positions obtained by both approaches are compared in the same image, illustrating how the PRH starts from an infeasible solution and obtains a feasible one.

7 | Conclusions

In this paper, we addressed the dispatching, positioning, and routing problem. This problem occurs after a fire is detected and consists of deciding which resources should be used to attack the fire, where to attack, and the route to follow from the base to that location. A layered network model was proposed that may be the base for approaches to other fire problems. Since resources move in a separate layer from fire, modeling their interaction is not straightforward. To address this, we proposed the concepts of suppression radius and safety radius. Striking a balance between keeping the model tractable and representing the interaction between resources and fire with appropriate detail is a key challenge that merits further exploration.

A MIP model and a heuristic were proposed. By design, comparison with MIP, and visualization, the heuristic seems to provide high-quality solutions. The computational experiments with different resolutions showed that it is possible to compromise the dimension of the landscape under study and the size of the network to obtain solutions in reasonable times. Scenarios with more than 1 000 fire nodes were solved in less than 15 s when the number of resources available resources is high and in less than 7 min when there are few available resources. The use of the Python framework pyO3F and, in particular, its interface with a geographic information system, proved to be of utmost relevance for visualization in debugging and testing.

The extension of the proposed approaches to support extended attack decisions is a natural follow-up to this work. In that case, routing with more than one attack position must be considered and heuristics conceived (since a practical application is pursued). Another relevant issue is to consider other indicators (in addition to fire growth), in particular, those related to fire containment (see Reference [14] for MIP models but no problem-specific solution methods).

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Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

References

1. K. Haynes, K. Short, G. Xanthopoulos, D. Viegas, L. M. Ribeiro, and R. Blanchi, “Wildfires and Wui Fire Fatalities,” in *Encyclopedia of Wildfires and Wildland-Urban Interface (WUI) Fires*, ed. S. L. Manzello (Springer, 2020), 1–16.
2. C. Eberle and O. Higuera Roa, “Technical Report: Mediterranean Wildfires,” 2022.
3. E. Marris, “Hawaii Wildfires: Did Scientists Expect Maui to Burn?,” *Nature* 620 (2023): 708–709.
4. M. A. Finney, “Fire Growth Using Minimum Travel Time Methods,” *Canadian Journal of Forest Research* 32, no. 8 (2002): 1420–1424.
5. A. Bergsten Mendes and F. Pereira e Alvelos, “Iterated Local Search for the Placement of Wildland Fire Suppression Resources,” *European Journal of Operational Research* 304, no. 3 (2023): 887–900.
6. Comando Nacional de Emergência e Proteção Civil, *Diretiva operacional nacional no. 2 - decr dispositivo especial de combate a incêndios rurais 2023*. Technical report, 2023.
7. D. L. Martell, “A Review of Operational Research Studies in Forest Fire Management,” *Canadian Journal of Forest Research* 12, no. 2 (1982): 119–140.
8. D. L. Martell, “A Review of Recent Forest and Wildland Fire Management Decision Support Systems Research,” *Current Forestry Reports* 1, no. 2 (2015): 128–137.
9. T. J. Duff and K. G. Tolhurst, “Operational Wildfire Suppression Modelling: A Review Evaluating Development, State of the Art and Future Directions,” *International Journal of Wildland Fire* 24, no. 6 (2015): 735–748.
10. J. P. Minas, J. W. Hearne, and J. W. Handmer, “A Review of Operations Research Methods Applicable to Wildfire Management,” *International Journal of Wildland Fire* 21, no. 3 (2012): 189–196.
11. M. Bashiri, E. Nikzad, A. Eberhard, J. Hearne, and F. Oliveira, “A Two Stage Stochastic Programming for Asset Protection Routing and a Solution Algorithm Based on the Progressive Hedging Algorithm,” *Omega* 104 (2021): 102480.
12. N. Altay and W. G. Green III, “Or/MS Research in Disaster Operations Management,” *European Journal of Operational Research* 175, no. 1 (2006): 475–493.
13. B. Granda, J. León, B. Vitoriano, and J. Hearne, “Decision Support Models and Methodologies for Fire Suppression,” *Firehouse* 6, no. 2 (2023): 37.
14. F. Alvelos, “Mixed Integer Programming Models for Fire Fighting,” in *International Conference on Computational Science and Its Applications* (Springer, 2018), 637–652.
15. E. J. Belval, Y. Wei, and M. Bevers, “A Mixed Integer Program to Model Spatial Wildfire Behavior and Suppression Placement Decisions,” *Canadian Journal of Forest Research* 45, no. 4 (2015): 384–393.
16. M. G. Harris, M. A. Forbes, and T. Taimre, “Logic-Based Benders Decomposition for Wildfire Suppression,” *Computers & Operations Research* 160 (2023): 106392.
17. J. Hof, P. N. Omi, M. Bevers, and R. D. Laven, “A Timing-Oriented Approach to Spatial Allocation of Fire Management Effort,” *Forest Science* 46, no. 3 (2000): 442–451.
18. W. Yu, D. B. Rideout, and T. B. Hall, “Toward Efficient Management of Large Fires: A Mixed Integer Programming Model and Two Iterative Approaches,” *Forest Science* 57, no. 5 (2011): 435–447.
19. E. J. Belval, Y. Wei, and M. Bevers, “A Stochastic Mixed Integer Program to Model Spatial Wildfire Behavior and Suppression Placement Decisions With Uncertain Weather,” *Canadian Journal of Forest Research* 46, no. 2 (2016): 234–248.

20. A. Bergsten Mendes and F. Pereira e Alvelos, "A Robust Optimisation Approach for the Placement of Forest Fire Suppression Resources," *International Transactions in Operational Research* 32, no. 3 (2025): 1312–1342.
21. A. Shahidi, R. Ramezani, and S. Shahparvari, "A Greedy Heuristic Algorithm to Solve a Vrp-Based Model for Planning and Coordinating Multiple Resources in Emergency Response to Bushfires," *Scientia Iranica* (2022).
22. M. Van Der Merwe, J. P. Minas, M. Ozlen, and J. W. Hearne, "A Mixed Integer Programming Approach for Asset Protection During Escaped Wildfires," *Canadian Journal of Forest Research* 45, no. 4 (2015): 444–451.
23. E. Nikzad and M. Bashiri, "A Two-Stage Stochastic Programming Model for Collaborative Asset Protection Routing Problem Enhanced With Machine Learning: A Learning-Based Matheuristic Algorithm," *International Journal of Production Research* 61, no. 1 (2023): 81–113.
24. E. J. Belval, Y. Wei, and M. Bevers, "Modeling Ground Firefighting Resource Activities to Manage Risk Given Uncertain Weather," *Forests* 10, no. 12 (2019): 1077.
25. E. W. Dijkstra, "A Note on Two Problems in Connexion With Graphs," *Numerische Mathematik* 1 (1959): 269–271.
26. R. K. Ahuja, T. L. Magnanti, and J. B. Orlin, *Network Flows: Theory, Algorithms, and Applications* (Prentice Hall, 1993).
27. Direção-Geral do Território, "Especificações técnicas da carta de uso e ocupação do solo (COS) de Portugal continental para 2018," Informação geográfica cedida pela Direção-Geral do Território, 2019, <https://www.dgterritorio.gov.pt/>.
28. R. C. Rothermel, "A mathematical model for predicting fire spread in wildland fuels," Intermountain Forest & Range Experiment Station, Forest Service, 1972, 115.
29. P. Fernandes, H. Gonçalves, C. Loureiro, et al., "Modelos de combustível florestal para Portugal," in *Actas do 6o Congresso Florestal Nacional*. SPCF, Lisboa, Portugal, 2009, 348–354.
30. P. Fernandes and C. Loureiro, *Modelos de combustível florestal para Portugal, documento de referência. Technical report*, 2021.
31. P. L. Andrews, "Current Status and Future Needs of the Behaveplus Fire Modeling System," *International Journal of Wildland Fire* 23, no. 1 (2013): 21–33.
32. P. L. Andrews, "The Rothermel Surface Fire Spread Model and Associated Developments: A Comprehensive Explanation. USDA Forest Service, Rocky Mountain Research Station. Technical report, General Technical Report RMRS-GTR-371. (Fort Collins, CO)," 2018.
33. Comissão técnica independente, Análise e apuramento dos factos relativos aos incêndios que ocorreram em Pedrogão Grande, Castanheira de Pera, Ansião, Alvaiázere, Figueiró dos Vinhos, Arganil, Góis, Penela, Pampilhosa da Serra, Oleiros e Sertão, entre 17 e 24 de junho de 2017, 2017.
34. OpenStreetMap contributors, "Planet dump," 2017, <https://planet.osm.org>, <https://www.openstreetmap.org>.